

Program overview

20-May-2025 12:54

Year 2024/2025
Organization Mechanical Engineering
Education Master Offshore and Dredging Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master ODE 2024			
Master Offshore & Dredging Engineering 2024			
MSc ODE - Track Offshore and Dredging Engineering			
MSc ODE 1st year Obligatory Courses (60 ECTS)			
MT44021	Motions and Loading of Structures in Waves	5	
OE44085	Fatigue in Offshore Structures	5	
OE44150	Marine Hydromechanics	5	
OE44170	Offshore Renewables Technologies	5	
OE44175	Structural Mechanics of Offshore Structures	5	
OE44180	Seabed Preparation and Installation Process	5	
OE44185	Numerical Methods for Offshore and Dredging Engineering	7	
OE44190	Offshore Research Project	7	
OE44195	Reflection for Offshore and Dredging Engineers	1	
OE44200	Offshore Development Project	5	
OE44205	Offshore Geotechnical Engineering	5	
OE44210	Offshore Design Project	5	
MSc ODE 2nd year			
MSc Offshore & Dredging Engineering 2nd year			
MSc ODE 2nd year suggested elective courses			
CEGM3000	Multidisciplinary Project (MDP)	15	
CIEM4302	Cold Regions Engineering	5	
IFM4040	Joint Interdisciplinary Project	15	
MT44030	Advanced Mechanics of Maritime Structures	5	
MT44090	Fluid - Structure Interaction in Maritime Structures	5	
OE44215	Marine Pipelines	5	
OE44220	ODE Research Exercise	5	
RO47001	Robot Dynamics & Control	5	
RO47002	Machine Learning for Robotics	5	
SC42001	Control System Design	5	
SC42015	Control Theory	6	
SC42056	Optimization for Systems and Control	3	
SET3055	Economics and Regulations of Sustainable Energy Systems	4	
TPM404B	Technology Entrepreneurship and Internationalization	5	
ODE Graduation Project (Obligatory 45 ECTS)			
OE54010-22	ODE Literature Research	10	
OE54035	ODE Thesis	35	
MSc ODE - Track Offshore Renewables Engineering (EWEM)			
MSc ODE Semester 1 - DTU			
MSC ODE Semester 2 - TUD			
MSC ODE Semester 2 - TUD Obligatory courses			
OE44085	Fatigue in Offshore Structures	5	
OE44185	Numerical Methods for Offshore and Dredging Engineering	7	
OE44190	Offshore Research Project	7	
OE44195	Reflection for Offshore and Dredging Engineers	1	
OE44200	Offshore Development Project	5	
OE44210	Offshore Design Project	5	
MSc ODE Semester 3 NTNU			
MSC ODE Semester 4 TU Delft or NTNU			
MSC ODE Semester 4 TU Delft or NTNU Graduation Project			
OE54015EWEM	EWEM Literature Research	15	
OE54030EWEM	EWEM MSc Thesis	30	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

Master ODE 2024	
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Director of Studies	Dr.ir. P. van der Male
Contact for students	For information, please contact msc-ode@tudelft.nl
Program Title	MSc Offshore & Dredging Engineering

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE - Track Offshore and Dredging Engineering
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Year	2024/2025
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MSc ODE 1st year Obligatory Courses (60 ECTS)

Introduction 1	This section outlines course requirements that are more or less universal - independent of ones further choices. The knowledge and skill conveyed by these activities forms the heart of the offshore engineering curriculum; nearly all of the subsequent offshore engineering courses build upon these. These courses are coloured green on the Offshore Engineering MSc Curriculum Diagram which may be downloaded from the website www.offshore.tudelft.nl. Detailed information on the core and specialisation courses is provided in the overview of courses. Also contact details of the teachers are given there.
Introduction 2	In below mentioned table all courses are mandatory with the following exception: * AES1730 is mandatory for students with a BSc Mechanical Engineering background * SC4026 or WB2105MT is mandatory for students with a BSc Civil Engineering background

MT44021	Motions and Loading of Structures in Waves	5
Responsible Instructor	Dr.ir. P. Naaijen	
Instructor	Dr.ir. P.R. Wellens	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	Motions and loading of structures is about the following topics. Part 1: Part 1 covers theory of linear excitation and linear motion response of floating structures in waves. Students learn how the linear motion transfer function (RAO) relating ship motion response to incoming wave is computed, including the numerical approach involved and how workability in a given wave spectrum can be assessed including extreme value statistics. In Part 2: Higher order potential theory and inclusion of non-linear effects in ship motions. Applications to motion of moored ships and to the determination of workability. The use of Computational Fluid Dynamics (CFD) for the determination of extreme loads on moored FPSO in extreme seas. One exercise accompanies this part.	
Course Contents	Floating structures: wave forces & motions, nonlinear problems, applications Participants completing this segment successfully will have a superior knowledge of and be able to predict the motion of floating bodies in the sea. They will be familiar with first order ship motions in irregular waves as well as drift forces, resulting from nonlinear phenomena. They can also apply this to applications such as station keeping and the determination of offshore workability. Part 1, instructed by Peter Naaijen, is about linear forces, motions and about potential coefficients. Part 2, instructed by Peter Wellens, is about nonlinear forces and the resulting motions.	
Study Goals	Participants who have successfully completed the course will be able to carry out computations at a superior knowledge level involving: - linear forces on and resulting motions of ships - nonlinear forces on and resulting ships motions - workability prediction.	
Education Method	Lectures, exercises and assistance sessions for exercises	
Literature and Study Materials	For Part 1, separate lecture notes by Peter Naaijen are available. As an additional reference the Lecture notes "Offshore Hydromechanics" by Journée, Massie and Huijsmans can be used. Exercises "Offshore Hydromechanics" by Journée and Massie.	
Prerequisites	All participants are required to have successfully completed a basic university level courses in dynamics (mass-spring-damper systems etc.), fluid mechanics and free surface waves before starting this course.	
Assessment	Exams: Both part 1 and part 2 provide a summative assessment (exercise) which is not graded. One graded written exam (summative) for Part 1 and Part 2 together at the end of the course. (In case on site examination will not be possible due to COVID19 related circumstances, online examination will be arranged.)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Aggregation of old courses MTM1410 and OE4630 part 2 and 3	
Department	ME Department Maritime & Transport Technology	

OE44085	Fatigue in Offshore Structures	5
Responsible Instructor	Dr.ir. J.H. den Besten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Offshore structures are exposed to cyclic mechanical loading: forces and moments, both environment (wind, waves) and service (operations, machinery) induced. The response in terms of stress, strain or energy is cyclic by nature accordingly, meaning fatigue, a cyclic loading & response induced local, progressive, structural damage mechanism is a governing limit state. To be able to assess the offshore structure integrity & longevity in this matter, course contents include: Fatigue fundamentals, i.e. physics and damage process phenomena in: materials at defect locations (stress concentrations at micro- and meso-scale); different sources and different stages in the fatigue & fracture process (crack initiation, growth and propagation, fracture). structures at notch locations (stress concentrations at macro-scale) as part of structural members or at the member connections; particular attention will be paid to arc-welded joints in that respect, since marine structures are predominantly assemblies consisting of plates, shells and tubular members. Macroscopic (notch) stress concentrations, distributions and intensities: as part of structural members. at (welded) structural member connections, consisting of a cyclic mechanical- and quasi-constant thermal residual part. Fatigue resistance: materials (plane geometries) structures (notched geometries: welded joints) at small-, large- and full-scale testing (measurement techniques) damage criterion life time relations in different ranges (low-, mid- and high-cycle fatigue); formulations and parameter estimates using regression analysis (least squares- and likelihood approaches). short- and long crack growth relations. Fatigue assessment concepts: damage criteria using an intact geometry parameter (nominal stress-, structural hot spot stress- and effective notch stress/ strain concept). damage criteria using a cracked geometry parameter (crack tip stress intensity- and equivalent stress concept). Particular attention will be paid to analytical formulations and numerical approaches; concept capabilities and limitations. Fatigue influence factors: material and geometry (strength, surface integrity, volume, welding improvement techniques), environment (temperature, corrosion), loading & response (min/max ratio, mean stress, amplitude variability (time domain analysis, cycle counting, spectral approaches, scatter diagrams, cumulative damage hypotheses), multi-axiality including proportionality (in-phase, out-of-phase). Fatigue reliability for (risk based) inspection and maintenance purposes: time dependent limit state functions, (joint) design variable statistics, level 1, 2 and 3 analysis, 1st and 2nd order analytical reliability methods and Monte Carlo simulation. Structure integrity & longevity: fatigue strength calculation and life time estimation for assessment and design using different (design) philosophies, concepts, rules, standards and guidance.	
Study Goals	Theory (fundamentals): to identify and describe the fatigue physics and damage process phenomena in materials & structures to establish and calculate the macroscopic fatigue characteristic structural response parameters to investigate fatigue resistance testing considerations and characterise damage criterion life time relations to model fatigue physics mathematically based on different damage criteria developed in the course of time to assess fatigue resistance consequences of relevant influence factors and incorporate models in fatigue damage criteria to examine 1st and 2nd order as well as Monte Carlo simulation based reliability methods at level 1, 2 and 3 for (risk based) inspection and maintenance purposes Validation (experiments): to evaluate dedicated fatigue testing techniques to analyse and process measurement data and establish fatigue resistance parameters Application (engineering): to identify and classify fatigue sensitive locations in structures to select an appropriate fatigue assessment concept to establish and improve marine structure integrity & longevity, i.e. to calculate the fatigue strength and corresponding life time estimate for analysis and design purposes	
Education Method	Each week 2 lectures will be provided, as well as 1 workshop. Material for each lecture will be provided for preparation purposes. The workshop can be used to work on the assignments. The teacher is available for consultation.	
Assessment	For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	

Tags	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations . Analysis Modelling Structural Mechanics
Department	ME Department Maritime & Transport Technology

OE44150	Marine Hydromechanics	5
Responsible Instructor	D. Fiscoletti	
Contact Hours / Week x/x/x/x	HC 4.0.0.0 W 2.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The main topics of this course are: - hydrostatics / floating stability - laws of fluid mechanics - potential flow - constant real (viscous) flow - free surface (regular & irregular) waves - wave spectra - wave forces on slender structures	
Study Goals	After successfully completing this course, you will be able to: - Identify the relevant hydromechanical principles of the ocean environment for the analysis of offshore systems - Formulate the hydromechanical principles of the ocean environment for the analysis of offshore systems - Determine the hydromechanical loading on fixed rigid offshore systems - Assess the implications of simplifications of hydromechanical principles and corresponding assumptions on the analysis of fixed rigid offshore systems	
Education Method	classes, practice workshops	
Literature and Study Materials	Offshore hydromechanics, third edition, by Massie, Journée and Huijsmans Handout for wave forces on slender structures	
Assessment	written exam on campus	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found at: https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses To participate in the exam, (timely) registration is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found at: https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme at: https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations	
Department	ME Department Maritime & Transport Technology	
Contact	The instructor for year 2024 will be Daniele Fiscoletti D.Fiscoletti@tudelft.nl	

OE44170	Offshore Renewables Technologies	5
Responsible Instructor	Dr. G. Lavidas	
Responsible Instructor	Dr.ir. A. Jarquin Laguna	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course introduces students to the offshore renewable resources of the ocean environment and the technologies for renewable energy conversion, with the main purpose of developing comprehension of the first principles of offshore renewables engineering. In addition, students are trained on transferable skills, relevant for their professional development.	
Study Goals	Afer this course, students should be able to: Identify the relevant renewable energy principles in the ocean environment for the analysis of offshore renewable energy systems. Formulate the renewable energy principles in the ocean environment for the analysis of offshore renewable energy systems. Determine the renewable energy potential in the ocean environment and be able to design and assess the working principles of offshore renewable energy systems. Assess the power production by different offshore renewable technologies, and compare the technological solutions. Handle constructive feedback, both in provision and reception, to define individual goals for professional development. Infer logical conclusions based on deductive, inductive and abductive reasoning, distinguishing basic elements of arguments and recognizing different types of arguments, cognitive biases and fallacies in reasoning. Build international, multi-cultural and multi-disciplinary collaborative environments, based on established considerations, and recognizing the value of diversity, equity and inclusion in teams. Identify societal and environmental impact of own work on the basis of principles of morality and integrity.	
Education Method	Lectures. Practical sessions. For more information please check the Brightspace page of the course.	
Assessment	Grading is based on a series of group assignments throughout the course	
Department	ME Department Maritime & Transport Technology	

OE44175	Structural Mechanics of Offshore Structures	5
Responsible Instructor	Prof.dr. A.V. Metrikine	
Instructor	Ir. J.W. Welleman	
Contact Hours / Week x/x/x/x	HC 0.6.0.0 W 0.2.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Your expected pre-knowledge for this course comprises: - Knowledge on computing statically determinate structures, including support reactions and force/moment diagrams. - Basic knowledge on computing structures using the Principle of Virtual Work. - Knowledge on kinematic, constitutive and equilibrium relations for basic deformation cases (extension, bending, torsion). - Knowledge on stress distributions on basic prismatic members and on deformations of trusses and frames. - Knowledge on computing statically indeterminate structures, including the force and displacement methods.	
Course Contents	This course treats the mechanics of slender structures. First, simple cases are treated in which analytical solutions of single elements are possible. The goal is to explain physics and the basics of the methods. Thereafter, the focus is shifted towards more design-type assignments in which the students apply the solution methods in more complex systems consisting of combined one-dimensional structural elements and plates together with complex boundary/interface conditions. The course covers: - Formulation of governing equations of one-dimensional and two-dimensional slender structures. - Statics and dynamics of cables, rods, shear beams and beams in bending. - Statics and dynamics of plates in bending. - Statics and dynamics of combinations of one-dimensional slender structures. In addition, students are trained on transferable skills, relevant for their professional development.	
Study Goals	After this course, you should be able to: 1. Identify the relevant mechanical principles in terms of one-dimensional slender structures and plates of offshore systems 2. Formulate the governing equations of motions of one-dimensional slender structures and plates 3. Determine the response of one-dimensional slender structures and plates to static and dynamic loads 4. Assess the implications of simplifications of physical principles and corresponding assumptions on the analysis of offshore & dredging engineering systems 5. Perform an analysis of personal characteristics in terms of core qualities, pitfalls, allergies and opportunities in relation to others, within the context of a professional environment 6. Evaluate alternative positions in discussions by taking up and evolving arguments brought in by others, recognizing moral values underlying own and alternative opinions, distinguishing personal opinions and factual judgments 7. Articulate a moral framework for the analysis of the impact of own work based on personal values and professional guiding principles	
Education Method	Lectures. Practical sessions. For more information please check BrightSpace.	
Computer Use	Maple, Python. Please check Brightspace for more detailed information.	
Course Relations	This course is part of the master programme Offshore & Dredging Engineering. The lectures overlap with the lectures of Unit 2: Mechanics of slender Structures of the Module CIEM5000 Track Base Structural Engineering of the master programme Civil Engineering. As a consequence, provided examples and exercises may not always explicitly relate to offshore and dredging engineering.	
Assessment	Study goals 1-4 will be assessed by means of a written exam, which requires a minimum grade of 5.8 for a pass. For students from (start) cohort 2024-25 and after: your performance on study goals 5-7 is expressed in an individual reflection portfolio, which will be assessed in the course OE44195.	
Enrolment / Application	For this course, enrolment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/regulations.	

OE44180	Seabed Preparation and Installation Process	5
Responsible Instructor	Dr.ir. R.L.J. Helmons	
Instructor	Dr.ir. S.M.S. Alhaddad	
Instructor	Dr.ir. G.H. Keetels	
Contact Hours / Week x/x/x/x	HC 0.4.0.0 W0.2.0.0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This course entails the seabed preparation processes that are typically encountered in dredging and marine construction. These seabed alterations are typically due to mechanical or hydraulic action, but could also happen due to natural processes occurring. The following content will be treated during the course: - Introduction to dredging engineering and dredging equipment - Intro to soil mechanics - Cutting mechanisms - Generic cutting model - Cutting of saturated sand - Clay cutting - Rock cutting - Settling velocity - Sedimentation & (high speed) erosion - Jetting in sand and clay - Breaching - Turbidity currents	
Study Goals	By the end of this course, the students will be able to: - Explain the fundamental principles that govern rapid deformation of (submerged) soils - Apply the physical principles that govern excavation processes in dredging engineering - Evaluate the factors that affect the excavation process, including soil conditions and equipment used (mechanical, hydraulic or hybrid). - Assess the strengths and limitations of different models and assumptions used to simplify the analysis of the excavation and seabed intervention processes.	
Education Method	Lectures, workshops	
Assessment	Assignments in small groups	
Department	ME Department Maritime & Transport Technology	

OE44185	Numerical Methods for Offshore and Dredging Engineering	7
Responsible Instructor	J.O. Colomes Gene	
Responsible Instructor	Dr.ir. G.H. Keetels	
Contact Hours / Week x/x/x/x	HC 0.0.0.2 W 0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is an integrated part of the master programme Offshore & Dredging Engineering. Your success in this course will therefore benefit from your active participation in this programme.	
Course Contents	This course concerns the numerical methods that can be used to analyse offshore & dredging engineering systems (structures and operations). During this course, you are trained to perform a numerical experiment, including the design of algorithms. Following through the different steps of study design, pre-analysis, data collection, post-analysis, validation and interpretation, you gain experience in drawing relevant conclusions and reflecting critically on the research activities performed. After a training on relevant numerical methods, covering both computational fluid and structural dynamics, you will conduct a numerical experiment using your self-developed algorithm in Python or Matlab. Following the FAIR principles, you will learn how to analyse and visualize your obtained data to draw scientifically relevant conclusions. The course offers multiple numerical assignments to conduct in small groups. You can express your motivated preference for a specific assignment through the individual reflection portfolio of the course OE44195. Your allocation to a specific assignment will be based on both your motivation and practical considerations.	
Study Goals	After this course, you should be able to: 1. Define a methodical approach to answer scientific research questions in the domain of offshore & dredging engineering, including the selection of appropriate methods and tools. 2. Conduct numerical experiments in a reproducible way, including the design of algorithms, using relevant numerical methodologies to analyze the working principles of offshore & dredging engineering systems. 3. Draw scientifically relevant conclusions (and the range of applicability thereof) from data obtained with numerical experiments using coherent interpretation techniques. 4. Disseminate data retrieved from numerical experiments required to answer research questions in a findable, accessible, interoperable and reusable manner. 5. Demonstrate academic integrity, implying commitment to honest and moral behaviour in an academic setting. 6. Evaluate collaboration and team performance in terms of clear targets, productivity, sharing of responsibilities and accountability, using constructive feedback. 7. Update the personal development plan in a reflective and responsive manner, based on monitoring of development using self-assessment and feedback from peers and supervisors	
Education Method	The course consists of both theoretical and practical lectures. During the former, you are trained in different theories and methods for conducting numerical research, whereas during the latter you can work under supervision on pre- and post-processing your numerical experiment.	
Course Relations	This course links to the course OE44195 Reflection for Offshore & Dredging Engineers. First of all, you can express your motivated preference for a numerical assignment through the individual reflection portfolio that is part of OE44195. Second, part of the learning objectives of this course will be assessed in the individual reflection portfolio of OE44195. This course intends to prepare you for your graduation project in the second year of the master programme. Active involvement in this course will be reflected positively in your literature review (OE55010-22) and Masters thesis (OE55035).	
Assessment	The course consists of group work that covers Study goals 1-4. The group work will be assessed based on a final report (67%), and a Q&A (33%). The Q&A will be preceded by a short presentation (10-15 minutes), and will result in an individual grade. In order to pass the course, both the report and the Q&A require a minimum grade of 5.8. The Q&A requires your active participation to receive a grade. In case the final report is graded lower than 5.8, your group has the opportunity to submit an updated report within the communicated time frame. Updated reports cannot receive a grade higher than 5.8. If your grade for the Q&A is lower than 5.8, you have the opportunity of a second individual Q&A. Your performance on study goals 5-7 is expressed in an individual reflection portfolio, which will be assessed in the course OE44195. Any deviation from the described assessment procedure requires approval from the Board of Examiners.	
Enrolment / Application	For this course, enrolment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/registration-courses-and-examinations/courses. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/regulations.	
Department	ME Department Maritime & Transport Technology	

OE44190	Offshore Research Project	7
Responsible Instructor	Dr. D. Wüthrich	
Responsible Instructor	Dr. C.L. Walters	
Responsible Instructor	Dr.ir. S.M.S. Alhaddad	
Contact Hours / Week x/x/x/x	HC 0.0.0.2 W 0.0.0.2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is an integrated part of the master programme Offshore & Dredging Engineering. Your success in this course will therefore benefit from your active participation in this programme.	
Course Contents	This course equips you with the critical skills necessary to conduct novel research in offshore and dredging engineering, with a focus on physical experimentation. You will follow a structured process, including conception of the experiments, design, pre-analysis, data collection, post-analysis, and validation. Through this, you will gain experience in interpreting collected data, drawing meaningful conclusions, and reflecting critically on your research activities. After training on relevant experimental methodologies, such as Pi theorem, scaling laws, measurement techniques and signal processing, as well as on safety in the labs, physical experiments will be carried out in a relevant lab. Following the FAIR principles, you will learn how to analyse and visualize your obtained data to draw scientifically relevant conclusions. The course offers multiple experimental set-ups to conduct experiments in small groups. You can express your motivated preference for a specific set-up through the individual reflection portfolio of the course OE44195. Your allocation to a specific setup will be based on both your motivation and practical considerations.	
Study Goals	By the end of this course, students should be able to: 1. Define a methodical approach to answer scientific research questions in the domain of offshore & dredging engineering, including the selection of appropriate methods and tools 2. Conduct physical experiments in a reproducible way using relevant experimental methodologies to analyze the working principles of offshore & dredging engineering systems 3. Draw scientifically relevant conclusions (and the range of applicability thereof) from experimental data using coherent interpretation techniques 4. Disseminate data retrieved from research required to answer research questions in a findable, accessible, interoperable and reusable manner 5. Demonstrate academic integrity, implying commitment to honest and moral behaviour in an academic setting 6. Evaluate collaboration and team performance in terms of clear targets, productivity, sharing of responsibilities and accountability, using constructive feedback 7. Update the personal development plan in a reflective and responsive manner, based on monitoring of development using self-assessment and feedback from peers and supervisors	
Education Method	The course consists of both theoretical and practical lectures. During the former, you will be trained in various theories and methods for conducting experimental research, whereas during the latter you can work under supervision on pre- and post-processing your experiment. Each group will conduct its physical experiments in one of the labs, the schedule of which will be made available at the start of the course. Successful participation in the course requires your active contribution to your lab experiments and extensive group work outside of lecture hours.	
Course Relations	This course links to the course OE44195 Reflection for Offshore & Dredging Engineers. First of all, you can express your motivated preference for an experimental set-up through the individual reflection portfolio that is part of OE44195. Second, part of the learning objectives of this course will be assessed in the individual reflection portfolio of OE44195. This course intends to prepare you for your graduation project in the second year of the master programme. Active involvement in this course will be reflected positively in your literature review (OE55010-22) and Master thesis (OE55035).	
Assessment	The course is assessed based on group work. To be assessed, your presence and active contribution during the lab experiment is required. The group work covers Study Goals 1-4. The group work will be assessed based on a final report (67%), and an oral Q&A session (33%). The Q&A will be preceded by a short presentation (10-15 minutes) and will result in an individual grade. To pass the course, both the report and the Q&A must receive a minimum grade of 5.8. The Q&A requires your active participation to receive a grade; students will be questioned individually on their report content and on topics covered during the course. In case the final report is graded lower than 5.8, your group will have the opportunity to submit an updated report by the end of the first September following the course. Updated reports cannot receive a grade higher than 5.8. If your grade for the Q&A is lower than 5.8, you have the opportunity of a second individual Q&A by the end of the first September following the course. Your performance on study goals 5-7 is expressed in an individual reflection portfolio, which will be assessed in the course OE44195. Any deviation from the described assessment procedure requires approval from the Board of Examiners.	
Enrolment / Application	For this course, enrolment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances, a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. To participate in this course, registration via MyTUDelft is not needed. The mandatory Q&A sessions will take place at the end of Q4, and the sign-ups for that will be done during the course. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme.	
Department	ME Department Maritime & Transport Technology	

OE44195	Reflection for Offshore and Dredging Engineers	1
Responsible Instructor	Dr.ir. P. van der Male	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	2 4	
Course Language	English	
Required for	Starting your thesis work (OE55010-22 and OE55035) requires a successful completion of this course.	
Course Contents	Your study is not just about grades and exams. You have chosen the master Offshore & Dredging Engineering to become a knowledgeable and skillful engineering professional. To this end, you will participate in all courses, and demonstrate your skills and understandings in exams and projects. Becoming an engineering professional involves more than demonstrating your performance in exams and projects. Each of us has her/his personal talents and preferences, and we all suffer from ineptitudes in certain areas. Being effective in the professional field requires a good level of self-awareness. On which aspects can you be a leader and for which domains can you better rely on the expertise of others? The course Reflection for Offshore & Dredging Engineers aims to open your eyes for all non-technical aspect that contribute to your success as engineering professional. Understanding your strengths and weaknesses helps you formulating goals for self-development and makes you an effective team player. Running in parallel with all your course work in the first year of your master, the course will help you sharpen your thoughts regarding the projects and elective courses you intend to follow. You are invited to motivate your preferred projects for the second semester and your selection of elective courses for the first quarter of the second year.	
Study Goals	Your reflection on your development as offshore & dredging engineer contributes to your proof that you meet the following exit qualifications of the programme: - Self-development: Develop competences and knowledge to function in varying professional environments. - Academic communication: Communicate professional and scientific judgments effectively. - Collaborative problem-solving: Collaborate in international, multi-cultural and multi-disciplinary teams. - Ethical Intelligence: Demonstrate the ability to critically assess the societal and environmental implications of their engineering solutions.	
Education Method	Throughout the first year of the programme you will work on an individual reflection portfolio. In this portfolio, you answer a prepared set of reflective questions for each subsequent quarter. By working on these questions during each quarter, you will complete the learning lines corresponding to the four learning objective of the course. In order to answer the questions, you reflect on your experiences during the courses of each subsequent quarter. Through BrightSpace, we will also provide inspirational materials, such as TedX presentations, YouTube videos and short texts and suggestions for useful books - which can help you during your reflections. You are of course invited to look for your own materials that help you shape your thoughts. Throughout your studies, you are invited to discuss the different topics with the lecturers of your courses and your fellow students.	
Literature and Study Materials	Your participation in the different courses of the programme should provide you with relevant experiences to reflect upon. Supporting study material and references are provided through BrightSpace. In addition, you are invited to explore further material in literature and on the internet.	
Assessment	While reflection can of course not be graded, proving your development is mandatory within the master Offshore & Dredging Engineering. Therefore, you need to submit an individual portfolio at two instances of your studies, at the end of Q2 and at the end of Q4, with the following corresponding deadlines: - Part 1 should be submitted latest on the Friday of week 2.8 - Part 2 should be submitted latest on the Friday of week 4.8 Both parts of the portfolio contain your answers to the eight reflective questions (four questions per quarter). In addition, there is a section in each part with your motivated choices for the following semester. In Part 1, you will motivate your choices regarding project work in the second semester and in Part 2 you will motivate your choices regarding elective courses in the first quarter of the second year. Based on your motivations, we can decide to honour your preferences. The default format of your portfolio is one written page per quarter for answering your questions (A4, font Arial, size 11) and maximum one page with your motivated choices per semester. The size of each part of your portfolio is therefore three pages at max. Nevertheless, you have the freedom to choose an alternative digital format of your portfolio. You could instead decide to submit your reflection as a voice recording, video or by means of a slide show, or any other format that fits your personal style and preference. The duration of any recording should be between 7-10 minutes per quarter. You can submit your portfolio in your personal folder on BrightSpace. Both parts of your reflection portfolio will be assessed as pass/not yet pass. Please know that your reflections as such can never be as 'not yet pass'. A 'not yet pass' would indicate that completeness and ownership of your reflections could be improved. In this case, you will have the opportunity to submit an updated version of your portfolio for second assessment. Based on your motivations, you will be assigned to projects in the second semester and approval of your selected will be decided upon. In addition, you will be invited for a mandatory interview at two instances throughout the year (end of Semesters 1 and 2). Latest one day before the interview, you should submit a draft version of your reflection portfolio. Further information regarding these interviews will be provided through BrightSpace. You can submit your draft portfolio in your personal folder on BrightSpace. Please note that failure to submit shifts the deadline to the next semester (e.g., from week 2.8 to week 4.8). Part 1 should always be submitted one semester before Part 2. Deviation from these regulations requires approval from the Board of Examiners.	
Enrolment / Application	In order to participate in the course, registration on BrightSpace is mandatory.	
Department	ME Department Maritime & Transport Technology	

OE44200	Offshore Development Project	5
Responsible Instructor	Dr.ir. R.L.J. Helmons	
Responsible Instructor	Dr.ir. A. Jarquin Laguna	
Instructor	Dr.ir. B.C. Ummels	
Instructor	Ir. A.C.M. van der Stap	
Contact Hours / Week x/x/x/x	HC 0.0.4.0 W 0.0.2.0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course will introduce the students to the technical aspects and detailed understanding of offshore developments (e.g. renewable energy farms, deep-sea mining, seabed preparations). The course builds on fundamental knowledge addressed in the first semester with project management techniques, as well as applied design methodologies. The students will participate in a realistic project teamwork, working on actual cases with stakeholder analysis, risk management assessment and modelling, including an overview of the project development of safe and reliable offshore technologies. In this course, students can choose from different development projects to participate in (currently two projects available).	
Study Goals	After this course, students should be able to: Define the design space of a new or modified offshore & dredging engineering development, based on a functional and requirements analysis, and accounting for the full life-cycle Select design solutions in a methodological and transparent manner, based on an analysis of the life-cycle performance in terms or functional, societal and economical value, including sustainability and circularity Perform risk analyses, accounting for health, safety and environmental risks, and including possible mitigation measures Create a plan for professional development, based on individual goals obtained from self-assessment and feedback from peers and supervisors Articulate assumptions, boundary conditions and limitations of (design) methods, distinguishing concept-based and application-based approaches, to put results, conclusions and recommendations into perspective Execute well-planned projects, based on principles of project management and time management Create morally adequate measures for reduction of impact or mitigation of adverse effects	
Education Method	Lectures and project teamwork	
Assessment	Grading based on a group assignment. Portfolio (to be evaluated as a part of the course Reflection for Offshore & Dredging Engineers)	
Department	ME Department Maritime & Transport Technology	

OE44205	Offshore Geotechnical Engineering	5
Responsible Instructor	Dr. E. Kementzetzidis	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course addresses relevant topics in Soil Mechanics and Geotechnical Engineering related to offshore energy renewable and oil&gas energy harvesting. The core subject is the analysis and design of the most common foundation systems for offshore structures. The course will start with an in-depth introduction to soil behaviour including the response of soils to environmental cyclic loading. Then, essential concepts about marine site investigation are discussed and followed by an introduction to soil-foundation-structure interaction effects. Then the course is brought to a conclusion by addressing the analysis/design of both shallow and deep offshore foundations.	
Study Goals	At the end of this course you will be able to: 1. recognise and describe the main features of offshore soil behaviour and soil-structure interaction under environmental loading; 2. describe the principles for planning offshore site surveys for soil characterisation purposes; 3. analyse/design the main shallow foundation systems according to standard analytical approaches 4. analyse/design offshore piles according to standard analytical approaches	
Education Method	The course material will be covered via in-class live sessions. Both fundamentals and practical applications will be covered via taught theory and practice assignments.	
Assessment	Assessment is fully through a written, open-book exam at the end of the education period. A similar resit will take place at the end of the following education period. Assignments will also be provided to the students.	
Department	ME Department Maritime & Transport Technology	

OE44210	Offshore Design Project	5
Responsible Instructor	Dr.ir. P. van der Male	
Responsible Instructor	Ir. J.S. Hoving	
Contact Hours / Week x/x/x/x	HC 0.0.4.0 W 0.0.2.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Successful participation in the following courses (or equivalent) is advised: - OE44150 Marine Hydromechanics - OE44170 Offshore Renewables Technologies - OE44175 Structural Mechanics of Offshore Structures - OE44180 Seabed Installation and Installation Process - OE44205 Offshore Geotechnical Engineering - MT44021 Motions and Loading of Structures in Waves	
Course Contents	The primary objective of this course is to enable students to produce conceptual engineering designs for offshore structures accounting for ultimate strength and fatigue criteria. During this course, students gain a thorough theoretical understanding of the design process, its structuring, available design methodologies and the philosophy behind the design codes. Throughout the course students work on a group design project, where each student team designs an offshore support structure from scratch up to a preliminary design by performing a full offshore support structure design-cycle. This design-cycle includes: I) the assembly of the structural configuration, II) the determination and assessment of the permanent, variable and environmental loads on the structure, III) the design and analysis of its foundation, and IV) the static and dynamic structural analysis of the structure based on ultimate strength and fatigue principles. By means of this group design project, students will learn to judge their design against ULS and FLS design criteria and be able to modify and improve their design based on static and dynamic structural analyses. Thereby, students will be able to analyse and evaluate the dimensioning and performance of structural components for strength and stability. Furthermore, students learn to predict how changing (parts of) their design will influence the corresponding loads and based on their insight in these loads, be able to defend their design choices. Moreover, students will learn what the practical implications of their design choices are for operations and other feasibility and life-cycle aspects. To reinforce the understanding of these aspects, the course program also includes a visit to an offshore structures fabrication yard. During this module, students will not only learn how to design an offshore structure, but they also learn how to collaborate effectively in a team and will further develop skills and competences to motivate their assumptions and provide argumentation of their design choices by accounting for relevant moral values, norms and principles.	
Study Goals	After a successful participation in this course, students are able to: 1. Define the design space of new or modified offshore & dredging engineering structures or operations, based on a functional and requirements analysis, and accounting for the full life-cycle. 2. Select design solutions in a methodological and transparent manner, based on an analysis of the life-cycle performance in terms of functional, societal and economical value, including sustainability and circularity. 3. Perform risk analyses, accounting for health, safety and environmental risks, and including possible mitigation measures. 4. Evaluate the limit states of designs of offshore structures, using an applicable safety philosophy (deterministic, probabilistic) based on valid design standards. 5. Create a plan for professional development, based on individual goals obtained from self-assessment and feedback from peers and supervisors. 6. Articulate assumptions, boundary conditions and limitations of (design) methods, distinguishing concept-based and application-based approaches, to put results, conclusions and recommendations into perspective. 7. Execute well-planned projects, based on principles of project management and time management. This course contributes to the final attainments of the Offshore & Dredging Engineering programme on Constructive Design (1-4), Self-Development (5), Academic Communication (6), Collaborative Problem-Solving (7) and Ethical Intelligence (8).	
Education Method	The education in this course includes problem-based lectures, workshops, a group design project and self-study. In total, this course has 8 contact hours per week, consisting of: - two 2-hour lectures (2x45 min) focused on comprehension and application of theory and methodology of offshore support structures design, and - a 4-hour workshop (4x45 mins) focused on problem analysis and evaluation, treating and discussing example problems, as well as to provide feedback and reflection on the group design project as well as allowing for discussion and Q&A regarding the group design project. The group design project is performed in groups and consists of 3 phases. Each phase comes with a formative assessment and feedback session per group in interaction with the lecturers and/or your peers, followed by a final deadline at the end of the teaching period at which the assignment is graded. The groups generally consist of 3 or 4 students, but the group size may be adapted depending on cohort size. Additionally, under reservation of unforeseen and/or external circumstances, the module includes an excursion to a fabrication yard where offshore support structures are constructed. Here note that the number of participants to this excursion is generally limited by the capacity at the yard and the excursion is therefore not a compulsory part of the module.	
Course Relations	This course is part of the MSc Offshore & Dredging Engineering (ODE) and the Offshore Engineering track of the European Wind Energy Master (EWEM-OE). The course runs parallel and in sync with the unit Design of Offshore Structures, embedded in the module CIEM4120 Offshore Structures, which is part of the track Hydraulic and Offshore Structures (HOS) of the MSc Civil Engineering. The course builds on the courses that are offered in the first and second quarters of the first year of the ODE curriculum.	
Literature and Study Materials	The lecture material for this course consists of the lecture slides, lecture notes, papers and chapters from relevant literature, and a basis for design document for the group design project, which will all be available through Brightspace.	
Assessment	Participants of this module are summatively assessed through a Group Design Project, worth 4EC, and an individual written exam, worth 1EC. The final grade for the course is the weighted average of the Group Design Project (80%) and the individual written exam (20%). To pass the course, participants must score a pass grade, i.e. at least a 6.0 (rounded to halves), on both the project and the exam. In addition, participation in the course allows students to work on an individual reflection portfolio, which is assessed within OE44195 Reflection in Offshore & Dredging Engineering (Study Goals 5-8).	

	Group Design Project: The final deadline for the Group Design Project is set at the end of the Q3 education period. If you fail to get a pass grade for the assignment, but the assignment grade is equal to or higher than a 5.0, you are allowed to improve your grade by submitting an addendum to your assignment report in the following education period, i.e. during Q4. Here, please note that the maximum grade that can be obtained after submitting an addendum to improve an insufficient grade is 6.0. Written exam: The regular individual written exam for this course is organised during the exam period of Q3, and a re-exam is organised in the exam period of Q4. The written exam is an open-book exam, meaning that any written or printed material is permitted during the exam sessions. If due to any unforeseen circumstances the written exam for this course may not be given in a plenary on-campus session, alternatively the written exam for this course may be given as an online open-book exam through Brightspace. To participate in the online exam you must be enrolled for this course in Brightspace. For further practicalities regarding any online examinations, you will be informed through Brightspace.
Permitted Materials during Tests	The written exam is an open-book exam, meaning that any books, slide print-outs, personal notes, i.e. any written or printed material is permitted during the exam sessions. However, to limit the amount of paper waste, a maximum of ten A4-pages (written or printed) are allowed during the exam. This rule does not include books. Additionally, it is allowed and advised to bring a scientific calculator to the exam. Any devices that allow for communication are not allowed, i.e. no laptops, pads or phones.
Enrolment / Application	For this course, enrollment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations.
Remarks	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/regulations. Through the past years, this course has seen several updates and improvements, and thereby updated course codes. Previous course codes are: OE4651, OE4651-12, OE4651-15, OE44095, OE44096 and OE44097.

Year

Organization

Education

2024/2025

Mechanical Engineering

Master Offshore and Dredging Engineering

MSc ODE 2nd year	
Introduction 1	The second master year consists of: Period 1: specialisation courses and/or electives and/or JIP Period 2: Literature Research Period 3 en 4: Thesis

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE 2nd year suggested elective courses

CEGM3000	Multidisciplinary Project (MDP)	15
Responsible Instructor	Dr.ir. S. de Vries	
Course Coordinator	Y. de las Heras	
Instructor	P. Taneja	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	The Multidisciplinary Project (MDP) is an MSc elective project course at the Faculty of Civil Engineering and Geosciences (CEG). All students that have started their masters programme in academic year 2024-2025 and onward are applicable to take the MDP, with the exception of the TIL, CME and AGP masters programmes. Students that started their master programme TIL or CME before academic year 2024-2025 can still start this project course if it is approved within their individual study programme. Note that taking the elective project MDP extracurricularly is still an option for TIL, CME and AGP students. In such cases the credits taken do not count for the 120 credits necessary for the given master programme. In the Multidisciplinary Project (MDP) students investigate a problem for a client in a team that consists of students from multiple tracks within the Faculty of Civil Engineering and Geosciences and develop a solution strategy or other advice. The client can be external or faculty staff. Students from other faculties can be involved, yet permission from the module coordinator is required for this. The faculty encourages students to perform a MDP abroad and to acquire a project themselves by reaching out to external organisations. Project topics are also offered via the online learning platform Brightspace. After acquisition, the MDP consists of three phases: Phase 1: Preliminary investigation. Students start with a problem analysis resulting in a problem statement and an objective. Coupled to that, a project strategy will be formulated. This strategy includes which methods will be used, which contribution can different disciplines provide to the project, which steps have been passed through successively, which information is still necessary, where that information can be found etc. In addition, the organization of the group will be set up in this phase. This phase ends with a presentation of the project proposal to the client and supervisors. Phase 2: Design and/or research. During this phase students will execute the strategy developed under phase 1 will and adjust it if needed. The work may exist of research and/or of developing design alternatives or solution alternatives. Alternatives may be worked out for the complete problem or for sub-problems. Half-way this phase the progress is presented in a midterm presentation. Phase 3: Round-off. In the round-off, students deliver the project results to the client and supervisors. First, a draft report is discussed with the supervisor and client, whereupon the definite version is made. The participants evaluate the project, both regarding the content and regarding the project process. Also a public final presentation is given.	
Study Goals	After completion of this module the students are able: MDP-LO1: Communicate project plans, intermediate and final results with a client and other relevant stakeholders in written and oral form. (ILO: i) MDP-LO2: Initiate, devise and plan project, in order to meet requirements, set by a client. (ILO: j) MDP-LO3: Monitor and manage a project in a team considering the client and other stakeholders involved. (ILO: j) MDP-LO4: Analyze a complex problem in the field of civil engineering, environmental engineering and/or geosciences. (ILO: j) MDP-LO5: Select and apply appropriate research and design methods to address this complex problem. (ILO: j) MDP-LO6: Understand the effect of diversity in disciplines, talents, skills, characters and cultures on a group process and address this effectively in a group process. (ILO: k) MDP-LO7: Apply working methods that stimulate combining and integrating knowledge and methods from different disciplines. (ILO: k)	
Education Method	MDP starts with a kick-off in which the steps of the project and methods for complex problem analysis and multidisciplinary working will be introduced. During the project students have regularly discussions with their supervisors and client. All these activities may be organized online to allow participation from abroad.	
Prerequisites	As of academic year 2023-24 Q2: students from must have obtained at least 45 ECTS. Mail the application form together with a study progress overview from each group member (all combined in one mail or one pdf) to studentoffice-esa-ceg@tudelft.nl. In the study progress overview EC's of parts of the modules can be used. This needs to be made clear by the student.	
Assessment	Formative assessment (feedback): Feedback from supervisors and client during presentations of project proposal and midterm. The draft report will be discussed with the group. Summative assessment: Students complete the module successfully when passing the assessment that consists of: 1. Quality written report and oral presentation (10%) 2. Quality research / final design as presented in written and oral report (50%) 3. Group process and multi disciplinarity (40%). A successful participation in the module, requires a minimum evaluation of 5.8 in all examinations	
Expected prior Knowledge	A large part of the first year of your program. See prerequisites.	
Academic Skills	During the project, students work on their skills regarding inquiry, collaboration, communication, integration of different	

Literature & Study Materials	perspectives, and reflection on personal functioning. Students will be offered handouts and web resources on problem analysis, working in diverse teams, project management, reporting a presenting. For problem analysis following material (or comparable resources) will be used together will dedicated material on Brightspace: https://ocw.tudelft.nl/courses/ngi-2/subjects/8-bonus-module-problem-demarcation-actor-analysis-skills-2/, (accessed 2-4-2021) For teamwork and project process following material (or comparable resources) will be used together will dedicated material on Brightspace: Schwartz, S. H. (2014). National culture as value orientations: Consequences of value differences and cultural distance. In Handbook of the Economics of Art and Culture (Vol. 2, pp. 547-586). Elsevier. https://www.belbin.com/about/belbin-team-roles https://www.mbtionline.com/ https://scrumguides.org/ https://online-learning.tudelft.nl/project-management-skills/ Students can use the resources below for report writing and presenting: http://eloquentscience.com/ https://faculty.washington.edu/heagerty/Courses/b572/public/StrunkWhite.pdf Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CIEM4302	Cold Regions Engineering	5
Responsible Instructor	Ir. J.S. Hoving	
Responsible Instructor	Dr.ir. H. Hendrikse	
Instructor	Dr.ir. H. Hendrikse	
Instructor	Ir. J.S. Hoving	
Contact Hours / Week x/x/x/x	4 (lectures) + 2 (workshop / question hour related to assignment)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Cold regions engineering relates to all engineering activities related to coping with low temperatures, specifically freezing temperatures, in the cold regions of our planet. The formal education of most civil engineers does not include an adequate understanding of the unique characteristics that work in these cold regions brings. Challenges include engineering problems related to soil mechanics, hydraulics, heat flow, materials and construction, among others, in cold environments. Therefore, this module addresses a broad range of cold regions engineering design and construction activities. Topics covered in this module are: - Physics and mechanics of sea, lake and river ice; - Ice growth, ice thermodynamics and icing; - Radiation, albedo, glacial energy budget and ice melt - Ice morphology, Ice features and ice regimes; - Cold regions structures and development; - Sea ice dynamics & remote sensing; - Ice actions on port, offshore and coastal structures; - Dynamic ice-structure interaction; Additional topics that may be shortly addressed depending on time available: - Material behavior at low temperatures; - Construction on ice and permafrost: foundations, roads and airfields; - Ships in ice;	
Study Goals	After completing this module, you will be able to: 1. Analyse the physics, mechanics and thermodynamics of ice. 2. Predict expected ice features and ice regimes in different cold regions. 3. Account for material behaviour in cold environments. 4. Classify cold region structures and developments. 5. Assess static ice-induced loads on port, coastal, marine and offshore structures. 6. Analyse and select optimal structure types and structural designs for performance in cold regions depending on local conditions. 7. Characterize and model dynamic ice-structure interaction.	
Education Method	Lectures, discussion of examples during practicals and exercises, as well as formative assignments.	
Assessment	Several formative group assignments on: 1. Ice material: Growth & Thermodynamics 2. Ice-structure interaction 3. Ice dynamics These assignments do not count towards the final grade, but serve as a means to test your knowledge and receive feedback on your understanding. Note furthermore that the size of the groups depends on the number of students in the course and submission deadlines for the group assignments are communicated through Brightspace. The course is concluded with an individual written open-book exam, where the pass grade is a minimum of 5,8.	
Special Information	This course is a new and updated version of the former course OE44115 Arctic Engineering.	
Expected prior Knowledge	Structural Mechanics / Dynamics (BSc)	
Academic Skills	Critical thinking, knowledge, design of experiments, design, writing, reporting	
Literature & Study Materials	Chapters from the ASCE Handbook for Cold Regions Engineering, Lecture notes on offshore engineering in Cold Regions, lecture slides, selection of scientific publications.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	Open book exam	
	Yes	

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should - be a 2nd-year master student; - have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. - be full time available (40 hours per week) in Q5; - meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman, VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (ILO: K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

The minimum grade for the summative assessment is ≥ 5.8 .

Process grade is 20%

The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

MT44030	Advanced Mechanics of Maritime Structures	5
Responsible Instructor	L. Pahlavan	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course 'Advanced Mechanics of Maritime Structures' focuses on the response of ship and offshore structures subject to torsion. The loading of maritime structures generally consists of normal and shear forces, as well as bending and torsion moments. Torsion becomes particularly important under staggered cargo loading conditions and when operating in quartering seas. These conditions introduce shear strain and stress, as well as normal strain and stress responses in the structural members. To effectively establish and calculate the torsional response of marine structures, the course covers the following subjects: Free Warping Torsion: Beam Stiffness and Strength Characteristics - Solid Cross-Sections: Saint-Venant theory - Hollow Cross-Sections: - Thin-Walled Open: Membrane analogy - Thick-Walled Closed - Thin-Walled Closed: Single-Cell (Bredt approximation) - Thin-Walled Closed: Multi-Cell - Thin-Walled Mixed (Hybrid) Free Warping Torsion: Beam Displacements - Solid Cross-Sections - Hollow Cross-Sections Constrained Warping Torsion: Beam Shear and Normal Stress - Hollow Cross-Sections: Thin-Walled Open - Hollow Cross-Sections: Thin-Walled Closed Generalised Torsion Equation - Hollow Cross-Sections: Thin-Walled Open - Hollow Cross-Sections: Thin-Walled Closed - Monitoring of Maritime Structures Subject to Torsion	
Study Goals	Theory (Fundamentals) - To model free warping torsion of beam-like structural members mathematically, and determine the stiffness, strength characteristics, and displacements for solid and hollow cross-sections. - To model constrained warping torsion of beam-like structural members mathematically, and calculate the induced shear and normal stresses for hollow cross-sections. - To establish the generalized beam torsion equation for hollow cross-sections. Application (Engineering) - To identify the torsion load-carrying members and their efficiency in marine structures at both local and global levels. - To calculate the torsion loading-induced response of marine structures for analysis and design purposes. - To familiarize with measurement and monitoring of the torsion loading-induced response of marine structures.	
Education Method	Lectures (using peer education) and assignments	
Assessment	Assignment reports (80%) and oral exam (20%)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Maritime & Transport Technology	

MT44090	Fluid - Structure Interaction in Maritime Structures	5
Responsible Instructor	Dr. A. Grammatikopoulos	
Instructor	Dr.ir. J.H. den Besten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	In order to define the topology and scantlings of body shape and structural members a fluid and structural dynamic analysis is required. The fluid dynamics analysis results include rigid body displacements and loading; input for the structural dynamics analysis, providing the response: body deflections and {strain, stress, energy}. In case the body deflections are small in comparison to the rigid body displacements, a one-way coupling (a hydro-structural analysis) is sufficient. However, if the body deflections become larger, the fluid dynamics; the loading, becomes non-negligibly affected in turn meaning a two-way coupling (a hydro-elastic analysis) is mandatory. Although the rigid body displacements hardly change, the response consequences can be significant as well as the limit state (i.e. buckling & ultimate strength, fatigue & fracture) effects. To be able to establish the hydro-structural and hydro-elastic response and limit state characteristics of marine (i.e. fixed and floating offshore) structures, course content includes: Fluid structure interaction phenomena, i.e. hydro-structural and hydro-elastic response aspects; one-way/two-way coupling formulations, for: 1. continuous wave loading: a. local, regional and global (whipping) structural response considerations b. frequency domain solutions c. testing (measurement techniques) 2. transient wave loading: impact (external: slamming, internal: sloshing) a. Von Karman and Wagner slamming model, including inertia and free vibration stage characteristics in case of wedge and cylinder water entry as well as wave impact b. local, regional and global (springing) structural response considerations c. time domain solutions: implicit/explicit numerical integration using Newmark and 4th order Runge-Kutta formulations d. testing (measurement techniques) Fluid dynamics: define and select body shape topology and scantlings using rigid body displacements and loading based on: 1. hydrodynamic boundary value problem formulations 2. local pressure- and global force/moment distributions Structural dynamics: define and select topology and scantlings of structural members using body deflections and stress/strain/energy response based on: 1. Euler/Timoshenko} beam element formulations 2. modal decomposition (frequency; natural (vibration) and wave (flexible) mode) 3. dry/wet characteristics 4. PS-SB symmetry (MB,V) and anti-symmetry (MB,H, MT) components Hydro-structural and hydro- elastic design criteria: loading and structure natural period aspects at local, regional and global level Limit state consequences: buckling & ultimate strength and fatigue & fracture Scaling (for global hydro-structural and hydro-elastic response testing purposes): 1. mass and stiffness properties and distributions 2. dry/wet natural frequency and natural (vibration) and wave (flexible) mode shape characteristics 3. backbone cross-section definition (hull girder similarity)	
Study Goals	Theory (fundamentals): to identify and describe the fluid structure interaction physics and phenomena for continuous/transient} wave loading and vortex induced vibrations at local/regional/global marine structural response level to establish hydrodynamic boundary value problem formulations as well as local pressure- and global force/moment distributions in order to define and select body shape topology and scantlings to establish Euler/Timoshenko beam formulations, modal decomposition, dry/wet characteristics and symmetry/anti-symmetry components in order to define and select topology and scantlings of structural members to model hydro-structural and hydro-elastic phenomena for continuous/transient wave loading and vortex induced vibrations mathematically based on different formulations developed over time Validation (experiments): to evaluate dedicated fluid structure interaction testing techniques to apply scaling laws for global hydro-structural and hydro-elastic response testing purposes w.r.t. mass and stiffness properties and distributions as well as dry/wet characteristics in order to establish backbone hull girder cross-section similarity to analyse and process measurement data and establish hydro-structural and hydro-elastic} design criteria Application (engineering): to identify and classify if a hydro-structural or hydro-elastic structural response calculation is sufficient	

Education Method	to calculate the hydro-structural and hydro-elastic response for analysis and design purposes in order to establish and improve the marine structure integrity & longevity; limit state consequences Lectures (blended)
Assessment	For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB, MAPLE and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .
Remarks	Course is programmed as a 2nd year MSc course.
Department	ME Department Maritime & Transport Technology

OE44215	Marine Pipelines	5
Responsible Instructor	Dr. K. Anupam	
Contact Hours / Week x/x/x/x	HC 0.0.4.0 W 0.0.2.0	
Course Contents	Marine Pipelines concentrates on different aspects of subsea pipeline design: Pipeline Design: The internal and structural design of pipelines for multi-phase (liquid gas) flows. Pipelines are dimensioned based upon flow in relation to properties of the transported material as well as capital expenditure and operating costs involved. The need for and means of providing thermal insulation is discussed including the measures of assuring flow in pipelines. Pipeline Route Selection: Routing of pipelines through the sea as well as their shore approaches are covered. Special attention is given to sub-sea tie-ins, pipeline and cable crossings, pipeline protection from fishing gear, shore approaches and pipeline trenching. The consequences of pipeline temperature changes and upheaval buckling are integral aspects of this topic as well as the on-bottom stability (pipelines on or in the sea bed). Pipelines Installation / Construction: This segment presents current and new technologies for the installation of pipelines in varying water depths, ranging from a few meters to depths measured in kilometers coupled with the role which installation plays in the design of a pipeline. Special attention is given to supporting finite element analysis (FEM) calculations, construction start-up, sea-bed lay-down, tie-ins and to welding technology. A classroom exercise is included as an integral aspect of the knowledge gained.	
Study Goals	Participants completing this course successfully will be able to function at an advanced to superior knowledge level productively and quantitatively in marine pipeline design teams.	
Education Method	Classroom/Online	
Assessment	Written exam + Case-study with video recorded presentation	

OE44220	ODE Research Exercise	5
Responsible Instructor	Dr.ir. S.M.S. Alhaddad	
Education Period	1 2 3 4	
Course Language	English	
Course Contents	In this course, students will undertake a research exercise to develop/enhance their research skills. The topic of the research exercise will be identified jointly by the student and their daily supervisor. The exercise may include various tasks such as literature review, data collection, data analysis, and hypothesis testing, all aimed at addressing specific research questions. This course will provide students with experience in conducting research, enabling them to apply theoretical knowledge in a practical context and gain a deeper understanding of how to conduct scientific research.	
Study Goals	Depending on the research exercise, the student will be able to conduct a set of the following research activities: 1. Survey the state-of-the-art knowledge pertaining to the problem at hand and identify relevant knowledge gaps. 2. Formulate research questions to be addressed within the allocated time for research. 3. Define a systematic approach to address the formulated research questions. 4. Execute the necessary tasks according to the defined approach. 5. Draw scientifically valid conclusions based on the findings.	
Education Method	Individual research exercise under daily supervision by a scientific staff member	
Assessment	Written report (weight = 90%) + Final presentation (weight = 10%). The research results should be documented in a report and subsequently presented to the daily supervisor and the responsible instructor.	
Department	ME Department Maritime & Transport Technology	

RO47001	Robot Dynamics & Control	5
Responsible Instructor	L. Peternel	
Instructor	B. Shyrokau	
Instructor	J.M. Prendergast	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the basics for deriving kinematic/dynamical models of a robot (such as a factory robotic arm, mobile robot, etc.) and for designing corresponding robot controllers. Students will learn how to derive the equations of motion and basic control equations themselves, and they will also learn how to use existing frameworks to do the same. THEORY SUBJECTS Introduction & Fundamentals -Course layout, instructors, grade scheme, etc. -Importance of the course and the big picture -Connection to real-world -Basic concepts/terms in robotics -Methods to describe robot pose -Numerical integration Recap of Control Theory -State-space model -Root locus -Bode plot -PID control -Concept of LQR and MPC control -Reference to online materials Endpoint Control for Robotic Manipulators -Big picture + motivation -Difference between position and torque-controlled robots -Endpoint position control (with a PID controller) -Endpoint impedance control (for position tracking tasks) -Endpoint force control (for force tracking tasks) Robotic Arm Kinematics -Deriving forward kinematics and Jacobian (for a 2-DoF planar arm) -Transformations between endpoint and joint space using Jacobian -Kinematic singularities and their solution -Kinematic redundancy and task priority control (for a 4-DoF planar arm) -Kinematics of parallel robots and differences compared to serial robots Robotic Arm Dynamics -Rigid-body dynamics model -Deriving dynamics model with Lagrange method for 2-DoF planar arm -Using the dynamics model in the control -Dynamic task priority control 3D Robotic Arm -Deriving 3D kinematics using the Denavit-Hartenberg convention -3D trajectory generation Planning & Motion Control -Orientation control -Stability analysis of the controller Kinematics & Dynamics of Mobile Robot / Automated Vehicle -Forward kinematic models -Wheel kinematic constraints -Robot kinematic constraints -Non-holonomic system -Derivation of linear bicycle model -Steady-state analysis Dynamics & Control of Mobile Robot / Automated Vehicle -Transient analysis -Frequency response -Path-following control PRACTICAL ASSIGNMENTS (individual) Assignment 1: The goal of this assignment is to learn how to control a 2-DoF robotic arm in Python to follow a prescribed trajectory using kinematic and dynamic equations. Assignment 2: The goal of this assignment is to learn how to control a multi-DoF robotic manipulator (3D kinematics, 3D trajectory, orientation control). Assignment 3: The goal of this assignment is to learn how to implement a linear bicycle model, and calculate and discuss primary handling characteristics, poles, stability analysis, and feedback control (PID).	
Study Goals	After completing this course, the students are able to:	

Education Method	-Derive the equations of motion for different types of robots -Create joint-level and endpoint-level controllers (including trajectory following) -Implement a simulation model in Python or Matlab -Analyse the designed control systems and reflect on the design choices -Lectures & interactive sessions (4 hours per week) -Practical assignments (homework)
Literature and Study Materials	Note: While attending lectures & interactive sessions is not mandatory, it is highly recommended. Assignments are not strictly mandatory, however, if you do not submit them you will lose the corresponding percentage of the final grade. That might mean that even if you pass the exam, you might not have a high enough final grade to pass the course. Study materials -Lecture slides -Papers, books, videos, etc. (links provided in Brightspace) Robotics textbooks (either one) -Lynch, K. M., & Park, F. C. (2017). Modern robotics. Cambridge University Press. -Craig, J. J. (2017). Introduction to robotics: mechanics and control. Pearson Education. -Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). Robotics: modelling, planning and control. Springer Science & Business Media. Control theory textbooks (either one) -Ogata, K. (2010). Modern control engineering. Prentice hall. -Dorf, R. C., & Bishop, R. H. (2001). Modern Control Systems: Solutions Manual (Vol. 12). Prentice Hall. Useful toolboxes -Corke, Peter. Robotics Toolbox for Python. -Foote, Tully. "tf: The transform library." 2013 IEEE Conference on Technologies for Practical Robot Applications (TePRA). IEEE, 2013.
Prerequisites	-Linear algebra (including matrix and vector operations) -Calculus -Linear system theory at the BSc level (e.g., simple differential equations, Laplace transforms) -Basic programming experience at the BSc level (Python/Matlab)
Assessment	Number of assessments: -Four (3 assignments, 1 exam), all based on individual work. -The exam will be a computer/digital exam in ANS on campus. Grade calculation: -If and only if the exam grade is 5.5 or higher, the grade will consist of 30% of the average score of the assignments and 70% of the exam grade. The final grade must be 6.0 or higher to pass the course. Assignment submission lateness criteria: <1 hour: -1 point <1 day: -2 points <2 days: -4 points <3 days: -6 points >3 days: no points given
Enrolment / Application	Note: Assignments and exam grades are only valid for the academic year they were obtained in. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department Contact	ME Department Cognitive Robotics For general course inquiries: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 1 and related content: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 2 and related content: Micah Prendergast (J.M.Prendergast@tudelft.nl) For Assignment 3 and related content: Barys Shyrokau (B.Shyrokau@tudelft.nl)

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr.ir. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning - Vanishing gradient problem, DropOut, Optimizers Advanced machine learning - Outlook: Recurrent neural networks, convolutional neural networks - Outlook: Data augmentation, AutoEncoder - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent	

	Systems" 3rd edition, Aurélien Geron
Prerequisites	Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation. Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design. Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination.	
Department	ME Department Delft Center for Systems and Control	

SC42015	Control Theory	6
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After successfully completing this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 mandatory assignments	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study. Written exam (individual, fully closed-book, no calculators nor formulas sheets allowed, counts for 70% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 30% of the final marks). To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75. Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next.	
Permitted Materials during Tests	none	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.	

SET3055	Economics and Regulations of Sustainable Energy Systems	4
Responsible Instructor	Dr. S.T.H. Storm	
Instructor	Prof.dr.ir. L.J. de Vries	
Instructor	G.O. Ndubuisi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	None	
Course Contents	This course is a general introduction to energy economics. In this course, you learn the essentials of how actors in the energy system make economic decisions, when markets work well and when they may fail, and what options the government has to remedy market failures. Specifically, the following topics are addressed: Introduction to energy (electricity) markets: perfect competition; oligopoly Rational (and irrational) market behaviour; ABM ? Investment decision making; LCOE Opportunity costing / discounting; NPV; IRR; discount rate Markets, (perfect) competition and market failure (esp. externalities & lock-in) Private costs, external cost and social cost; empirical evidence (related to energy) LCOE; discounting; private discount rate; IRR and NPV Social cost-benefit analysis; social discount rate	
Study Goals	After this course students are able to: evaluate and analyse the performance of energy (electricity) markets; identify cases of market failures and design remedial action to rectify these market failures; estimate the LCOE and calculate the NPV and IRR of an electricity investment project; contrast private and social costs of (renewable) electricity generation; assess technological lock-in in energy generation; and appraise the role of public policy in energy markets and in energy technology innovation.	
Education Method	This course consists of lectures. Parallel to the course a simulation game will be played by the students.	
Literature and Study Materials	To be announced.	
Assessment	The final grade of the course consists of the following components: - Individual computer exam (80%) - Report on the simulation game (20%) Students should score at least 6 for both parts. All marks will be expressed on a scale of 1 to 10. Final grade calculation: $(0.8 * \text{computer exam}) + (0.2 * \text{report})$ In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Computer exam: resit computer exam - Report on the simulation: resubmission of the report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Elective	Yes	
Co-Instructor	G.O. Ndubuisi	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	J. Schmutzler	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Elective	Yes	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

ODE Graduation Project (Obligatory 45 ECTS)
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OE54010-22	ODE Literature Research	10
Responsible Instructor	Dr.ir. P. van der Male	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This course serves as preparation for your master graduation project (OE54035). You are expected to do a review of the available literature for your intended topic of graduation, which result in the framework for your research. The report should consist of the following components: - A background description, or a description of the field. - A general problem statement. Why further research for this field is needed? - An overview of the available relevant literature. What research have been done? What problems have been considered? What are the recent developments? Other sources you could use are interviews, discussions with an expert panel, simple benchmark calculations etc. - A specific research objective, research question or hypothesis. What should be the outcome of your research? - A description of your research approach. How do you intend to reach your objective? For your literature research you should make use of recognized scientific sources. Correct, consistent and complete referencing in accordance with an accepted academic style is expected. It is possible to use oral sources, for instance by means of expert interviews. In this case, written documentations of the interview should be added as an appendix to the literature review. Online search engines and generative AI tools can be helpful in collecting information and improving your writing style. Using such tools does not take away your responsibility for the content of your report. A summary of your literature review can serve as the introduction of your graduation report. The procedure to find a topic for your literature review and graduation is as follows: - Contact the professor, or other academic staff members, who works on a topic of your interest to see if there is a project available that you might like. TU Delft is frequently contacted by problem owners for such projects, but this is not guaranteed. In a discussion you can both find out what subject interests you and if there is a project available that relates to that subject. In most cases there is a rough problem description available of about half a page (A4). If not, most academic staff members can tell you where to search or who to contact to help find a suitable project. - Another option is to contact a problem owner (company/PhD student/Staff Member) yourself (e.g. through your own network). If they agree to take you as a graduation student on a subject you find interesting, you should write a short description of about half a page (A4) to discuss with the specialization prof. This is usually sufficient to check if a problem is a viable subject to graduate on. - Based on the initial description and discussions with all involved parties, you should extend the description to a more detailed Plan of Approach that ensures that both you and your supervisory team agree on what you will do. If all parties agree with your Plan of Approach the initial phase is concluded and it is time to set the date for a kick-off meeting. Usually at this point, you should also be assigned a daily supervisor (the problem owner) and an academic supervisor (TU Delft). It is advised to start about 3 months in advance with these preparations.	
Study Goals	By the end of this course, you should be able to: - Search the literature systematically; - Formulate a research question; - Select representative scientific sources from several (for example, economic, ethical, environmental, medical, social, health) perspectives relevant to the assignment; - Draw conclusions related to the research problem and give recommendations towards new research opportunities; - Write a literature review in academic English; - Demonstrate a scientific attitude; - Manage your own learning process (including time management).	
Education Method	Self study	
Prerequisites	In accordance with Article 24 of the Rules and Guidelines of the Board of Examiners (RGBE), you can only be admitted to the masters thesis project if at least 54 EC of the first year of this master have been completed. The RGBE can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/regelingen-reglementen In addition, the course OE44195 has to be completed successfully, before a master's thesis project can be commenced. Make sure that all your courses are registered correctly in Osiris by means of a Form 2. In case any changes in your programme have taken place, you can submit a change form (Form 3). Form 2 and 3 can be submitted for approval to the ODE programme coordination (MSc-ODE@tudelft.nl). Both Form 2 and 3 can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren Before starting your thesis project, you need to submit your Form 4 to the ODE programme coordination (MSc-ODE@tudelft.nl). Form 4 can be found on the BrightSpace page of OE54022-10 and OE54035.	
Assessment	The assessment is based on the Rubric MSc Literature Research available in the ME student portal: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Maritime & Transport Technology	

OE54035	ODE Thesis	35
Responsible Instructor	Dr.ir. P. van der Male	
Contact Hours / Week x/x/x/x	0/x/x/x	
Course Contents	This course concerns the final part of the master programme Offshore & Dredging Engineering and follows on completion of the course OE54010-22 Literature Review (or the Specialization Project for EWEM-OE students). During the MSc thesis project, you will work individually on the execution of your experiment (physical, numerical, or other) that you designed to answer your research question. You will collect and analyze the resulting data, draw scientifically relevant conclusions and provide recommendations for further research. You will perform your activities under the guidance of a daily supervisor (problem owner, as defined under OE54010-22) and a academic supervisor (who could be the problem owner). Throughout your project, you will organize meetings with your supervisory team, during which you present your progress and bring up points for discussion. These meetings serve as an opportunity to receive feedback as well. The scheduling of the meetings, as well as the organization of the feedback process should be discussed with your supervisory team. The total duration of the MSc thesis project is approximately 25 weeks. In consultation with your supervisory team, it is possible to deviate from this. At the end of the 25 weeks, you submit a final draft of your thesis report and organize a so-called greenlight meeting. During this meeting, you present your work to your committee, who then decides whether you are ready to graduate (i.e. your work can be graded as 6.0 or higher). If the level of your work is insufficient, you will have a second opportunity to schedule a green light meeting, under the conditions agreed upon with your supervisory team. You also have the right to request feedback to improve your work towards the defense ceremony, even if a green light is granted. After receiving a green light, you can schedule your defence date and submit a Form 5 to the Study Programme Administration (spa-me@tudelft.nl), which is available at the following link: https://www.tudelft.nl/en/student/me-student-portal/education/related/student-forms/msc-forms/. At the defence ceremony, you will present your work to the public (family, friends, peers) and the assessment committee. The public and the committee are requested to ask questions about your work. After the presentation and the Q&A, the assessment committee will decide upon your final grade, which will be communicated in public. If you undertakes the graduation project at a company, a graduation agreement should be signed. In this respect, the template of the TUD-ME graduation agreement should be used, which is available at the following link: https://www.tudelft.nl/en/student/me-student-portal/education/related/student-forms/msc-forms/.	
Study Goals	The MSc Thesis project concerns the final component of the Master's programme. In accordance with the assessment rubric (see assessment), you should demonstrate that you are able to: - independently collect, process and integrate theory from different fields or sources - rigorously prove a well-structured development of new theory through the use of advanced mathematical, numerical or experimental methods, or - rigorously validate a well-structured development of a new design, or design method, through the use of advanced design methods and analyses - demonstrate a very high level of aptitude for the research or design area - produce results that can be published in a journal publication or can be incorporated in the intended application - write an excellent report in terms of contents, structure, referencing and clarity - provide a very clear presentation, very well organized, very good selection of information, very good eye contact, very clear voice, very clear answers - offer new insights during discussion, in depth argumentation, leading to a very interesting scientific meeting, detailed argumentation for all questions - communicate in excellent English writing and speaking skills - organize regular meetings, discussions on relevant and challenging topics initiated by the student; excellent time, excellent content - act as project manager of the research project, initiated new related projects and initiatives, report was written independently, excellent time planning - demonstrate exceptional practical (experimental/computer/data acquisition) skills; is always aware of safety issues, very careful and precise - demonstrates excellent critical attitude towards own results, literature and specialists - actively seek for feedback to improve her/himself	
Education Method	The MSc thesis project concerns an individual project. Progress and feedback can be discussed with the supervisory team.	
Prerequisites	In accordance with Article 24 of the Rules and Guidelines of the Board of Examiners (RGBE), you can only be admitted to the masters thesis project if at least 54 EC of the first year of his master have been completed. The RGBE can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/regelingen-reglementen In addition, the course OE44195 has to be completed successfully, before a master's thesis project can be commenced. Make sure that all your courses are registered correctly in Osiris by means of a Form 2. In case any changes in your programme have taken place, you can submit a change form (Form 3). Form 2 and 3 can be submitted for approval to the ODE programme coordination (MSc-ODE@tudelft.nl). Both Form 2 and 3 can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren Before starting your thesis project, you need to submit your Form 4 to the ODE programme coordination (MSc-ODE@tudelft.nl). Form 4 can be found on the BrightSpace page of OE54022-10 and OE54035.	
Assessment	For the assessment, use is made of the ME Master Thesis Grading Rubric, which can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren The rubric distinguishes three main components: - content - communication - process The content is assessed based on the final report. Communication is assessed based on the report, the final presentation and the Q&A. The process is assessed based on the communication between the student and the supervisory team throughout the project. The assessment is done by the assessment committee, which consists of the academic supervisor and an external academic member. This external member cannot assess the process. In case you are eligible for cum laude, an additional member of the assessment committee is required. For the requirements regarding cum laude, please consider Articles 30 and 30A of the RGBE (https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/regelingen-reglementen)	

Year

Organization

Education

2024/2025

Mechanical Engineering

Master Offshore and Dredging Engineering

MSc ODE - Track Offshore Renewables Engineering (EWEM)	
Program Title	The involvement of TU Delft in ODE track Offshore Renewables Engineering (joint-degree European Wind Energy Master track Offshore Engineering) will be discontinued. See the Addendum TER ODE for more information

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE Semester 1 - DTU

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSC ODE Semester 2 - TUD

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSC ODE Semester 2 - TUD Obligatory courses

OE44085	Fatigue in Offshore Structures	5
Responsible Instructor	Dr.ir. J.H. den Besten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Offshore structures are exposed to cyclic mechanical loading: forces and moments, both environment (wind, waves) and service (operations, machinery) induced. The response in terms of stress, strain or energy is cyclic by nature accordingly, meaning fatigue, a cyclic loading & response induced local, progressive, structural damage mechanism is a governing limit state. To be able to assess the offshore structure integrity & longevity in this matter, course contents include: Fatigue fundamentals, i.e. physics and damage process phenomena in: materials at defect locations (stress concentrations at micro- and meso-scale); different sources and different stages in the fatigue & fracture process (crack initiation, growth and propagation, fracture). structures at notch locations (stress concentrations at macro-scale) as part of structural members or at the member connections; particular attention will be paid to arc-welded joints in that respect, since marine structures are predominantly assemblies consisting of plates, shells and tubular members. Macroscopic (notch) stress concentrations, distributions and intensities: as part of structural members. at (welded) structural member connections, consisting of a cyclic mechanical- and quasi-constant thermal residual part. Fatigue resistance: materials (plane geometries) structures (notched geometries: welded joints) at small-, large- and full-scale testing (measurement techniques) damage criterion life time relations in different ranges (low-, mid- and high-cycle fatigue); formulations and parameter estimates using regression analysis (least squares- and likelihood approaches). short- and long crack growth relations. Fatigue assessment concepts: damage criteria using an intact geometry parameter (nominal stress-, structural hot spot stress- and effective notch stress/ strain concept). damage criteria using a cracked geometry parameter (crack tip stress intensity- and equivalent stress concept). Particular attention will be paid to analytical formulations and numerical approaches; concept capabilities and limitations. Fatigue influence factors: material and geometry (strength, surface integrity, volume, welding improvement techniques), environment (temperature, corrosion), loading & response (min/max ratio, mean stress, amplitude variability (time domain analysis, cycle counting, spectral approaches, scatter diagrams, cumulative damage hypotheses), multi-axiality including proportionality (in-phase, out-of-phase). Fatigue reliability for (risk based) inspection and maintenance purposes: time dependent limit state functions, (joint) design variable statistics, level 1, 2 and 3 analysis, 1st and 2nd order analytical reliability methods and Monte Carlo simulation. Structure integrity & longevity: fatigue strength calculation and life time estimation for assessment and design using different (design) philosophies, concepts, rules, standards and guidance.	
Study Goals	Theory (fundamentals): to identify and describe the fatigue physics and damage process phenomena in materials & structures to establish and calculate the macroscopic fatigue characteristic structural response parameters to investigate fatigue resistance testing considerations and characterise damage criterion life time relations to model fatigue physics mathematically based on different damage criteria developed in the course of time to assess fatigue resistance consequences of relevant influence factors and incorporate models in fatigue damage criteria to examine 1st and 2nd order as well as Monte Carlo simulation based reliability methods at level 1, 2 and 3 for (risk based) inspection and maintenance purposes Validation (experiments): to evaluate dedicated fatigue testing techniques to analyse and process measurement data and establish fatigue resistance parameters Application (engineering): to identify and classify fatigue sensitive locations in structures to select an appropriate fatigue assessment concept to establish and improve marine structure integrity & longevity, i.e. to calculate the fatigue strength and corresponding life time estimate for analysis and design purposes	
Education Method	Each week 2 lectures will be provided, as well as 1 workshop. Material for each lecture will be provided for preparation purposes. The workshop can be used to work on the assignments. The teacher is available for consultation.	
Assessment	For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	

Tags	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .
	Analysis Modelling Structural Mechanics
Department	ME Department Maritime & Transport Technology

OE44185	Numerical Methods for Offshore and Dredging Engineering	7
Responsible Instructor	J.O. Colomes Gene	
Responsible Instructor	Dr.ir. G.H. Keetels	
Contact Hours / Week x/x/x/x	HC 0.0.0.2 W 0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is an integrated part of the master programme Offshore & Dredging Engineering. Your success in this course will therefore benefit from your active participation in this programme.	
Course Contents	This course concerns the numerical methods that can be used to analyse offshore & dredging engineering systems (structures and operations). During this course, you are trained to perform a numerical experiment, including the design of algorithms. Following through the different steps of study design, pre-analysis, data collection, post-analysis, validation and interpretation, you gain experience in drawing relevant conclusions and reflecting critically on the research activities performed. After a training on relevant numerical methods, covering both computational fluid and structural dynamics, you will conduct a numerical experiment using your self-developed algorithm in Python or Matlab. Following the FAIR principles, you will learn how to analyse and visualize your obtained data to draw scientifically relevant conclusions. The course offers multiple numerical assignments to conduct in small groups. You can express your motivated preference for a specific assignment through the individual reflection portfolio of the course OE44195. Your allocation to a specific assignment will be based on both your motivation and practical considerations.	
Study Goals	After this course, you should be able to: 1. Define a methodical approach to answer scientific research questions in the domain of offshore & dredging engineering, including the selection of appropriate methods and tools. 2. Conduct numerical experiments in a reproducible way, including the design of algorithms, using relevant numerical methodologies to analyze the working principles of offshore & dredging engineering systems. 3. Draw scientifically relevant conclusions (and the range of applicability thereof) from data obtained with numerical experiments using coherent interpretation techniques. 4. Disseminate data retrieved from numerical experiments required to answer research questions in a findable, accessible, interoperable and reusable manner. 5. Demonstrate academic integrity, implying commitment to honest and moral behaviour in an academic setting. 6. Evaluate collaboration and team performance in terms of clear targets, productivity, sharing of responsibilities and accountability, using constructive feedback. 7. Update the personal development plan in a reflective and responsive manner, based on monitoring of development using self-assessment and feedback from peers and supervisors	
Education Method	The course consists of both theoretical and practical lectures. During the former, you are trained in different theories and methods for conducting numerical research, whereas during the latter you can work under supervision on pre- and post-processing your numerical experiment.	
Course Relations	This course links to the course OE44195 Reflection for Offshore & Dredging Engineers. First of all, you can express your motivated preference for a numerical assignment through the individual reflection portfolio that is part of OE44195. Second, part of the learning objectives of this course will be assessed in the individual reflection portfolio of OE44195. This course intends to prepare you for your graduation project in the second year of the master programme. Active involvement in this course will be reflected positively in your literature review (OE55010-22) and Masters thesis (OE55035).	
Assessment	The course consists of group work that covers Study goals 1-4. The group work will be assessed based on a final report (67%), and a Q&A (33%). The Q&A will be preceded by a short presentation (10-15 minutes), and will result in an individual grade. In order to pass the course, both the report and the Q&A require a minimum grade of 5.8. The Q&A requires your active participation to receive a grade. In case the final report is graded lower than 5.8, your group has the opportunity to submit an updated report within the communicated time frame. Updated reports cannot receive a grade higher than 5.8. If your grade for the Q&A is lower than 5.8, you have the opportunity of a second individual Q&A. Your performance on study goals 5-7 is expressed in an individual reflection portfolio, which will be assessed in the course OE44195. Any deviation from the described assessment procedure requires approval from the Board of Examiners.	
Enrolment / Application	For this course, enrolment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/registration-courses-and-examinations/courses. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/regulations.	
Department	ME Department Maritime & Transport Technology	

OE44190	Offshore Research Project	7
Responsible Instructor	Dr. D. Wüthrich	
Responsible Instructor	Dr. C.L. Walters	
Responsible Instructor	Dr.ir. S.M.S. Alhaddad	
Contact Hours / Week x/x/x/x	HC 0.0.0.2 W 0.0.0.2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is an integrated part of the master programme Offshore & Dredging Engineering. Your success in this course will therefore benefit from your active participation in this programme.	
Course Contents	This course equips you with the critical skills necessary to conduct novel research in offshore and dredging engineering, with a focus on physical experimentation. You will follow a structured process, including conception of the experiments, design, pre-analysis, data collection, post-analysis, and validation. Through this, you will gain experience in interpreting collected data, drawing meaningful conclusions, and reflecting critically on your research activities. After training on relevant experimental methodologies, such as Pi theorem, scaling laws, measurement techniques and signal processing, as well as on safety in the labs, physical experiments will be carried out in a relevant lab. Following the FAIR principles, you will learn how to analyse and visualize your obtained data to draw scientifically relevant conclusions. The course offers multiple experimental set-ups to conduct experiments in small groups. You can express your motivated preference for a specific set-up through the individual reflection portfolio of the course OE44195. Your allocation to a specific setup will be based on both your motivation and practical considerations.	
Study Goals	By the end of this course, students should be able to: 1. Define a methodical approach to answer scientific research questions in the domain of offshore & dredging engineering, including the selection of appropriate methods and tools 2. Conduct physical experiments in a reproducible way using relevant experimental methodologies to analyze the working principles of offshore & dredging engineering systems 3. Draw scientifically relevant conclusions (and the range of applicability thereof) from experimental data using coherent interpretation techniques 4. Disseminate data retrieved from research required to answer research questions in a findable, accessible, interoperable and reusable manner 5. Demonstrate academic integrity, implying commitment to honest and moral behaviour in an academic setting 6. Evaluate collaboration and team performance in terms of clear targets, productivity, sharing of responsibilities and accountability, using constructive feedback 7. Update the personal development plan in a reflective and responsive manner, based on monitoring of development using self-assessment and feedback from peers and supervisors	
Education Method	The course consists of both theoretical and practical lectures. During the former, you will be trained in various theories and methods for conducting experimental research, whereas during the latter you can work under supervision on pre- and post-processing your experiment. Each group will conduct its physical experiments in one of the labs, the schedule of which will be made available at the start of the course. Successful participation in the course requires your active contribution to your lab experiments and extensive group work outside of lecture hours.	
Course Relations	This course links to the course OE44195 Reflection for Offshore & Dredging Engineers. First of all, you can express your motivated preference for an experimental set-up through the individual reflection portfolio that is part of OE44195. Second, part of the learning objectives of this course will be assessed in the individual reflection portfolio of OE44195. This course intends to prepare you for your graduation project in the second year of the master programme. Active involvement in this course will be reflected positively in your literature review (OE55010-22) and Master thesis (OE55035).	
Assessment	The course is assessed based on group work. To be assessed, your presence and active contribution during the lab experiment is required. The group work covers Study Goals 1-4. The group work will be assessed based on a final report (67%), and an oral Q&A session (33%). The Q&A will be preceded by a short presentation (10-15 minutes) and will result in an individual grade. To pass the course, both the report and the Q&A must receive a minimum grade of 5.8. The Q&A requires your active participation to receive a grade; students will be questioned individually on their report content and on topics covered during the course. In case the final report is graded lower than 5.8, your group will have the opportunity to submit an updated report by the end of the first September following the course. Updated reports cannot receive a grade higher than 5.8. If your grade for the Q&A is lower than 5.8, you have the opportunity of a second individual Q&A by the end of the first September following the course. Your performance on study goals 5-7 is expressed in an individual reflection portfolio, which will be assessed in the course OE44195. Any deviation from the described assessment procedure requires approval from the Board of Examiners.	
Enrolment / Application	For this course, enrolment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances, a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. To participate in this course, registration via MyTUDelft is not needed. The mandatory Q&A sessions will take place at the end of Q4, and the sign-ups for that will be done during the course. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme.	
Department	ME Department Maritime & Transport Technology	

OE44195	Reflection for Offshore and Dredging Engineers	1
Responsible Instructor	Dr.ir. P. van der Male	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	2 4	
Course Language	English	
Required for	Starting your thesis work (OE55010-22 and OE55035) requires a successful completion of this course.	
Course Contents	Your study is not just about grades and exams. You have chosen the master Offshore & Dredging Engineering to become a knowledgeable and skillful engineering professional. To this end, you will participate in all courses, and demonstrate your skills and understandings in exams and projects. Becoming an engineering professional involves more than demonstrating your performance in exams and projects. Each of us has her/his personal talents and preferences, and we all suffer from ineptitudes in certain areas. Being effective in the professional field requires a good level of self-awareness. On which aspects can you be a leader and for which domains can you better rely on the expertise of others? The course Reflection for Offshore & Dredging Engineers aims to open your eyes for all non-technical aspect that contribute to your success as engineering professional. Understanding your strengths and weaknesses helps you formulating goals for self-development and makes you an effective team player. Running in parallel with all your course work in the first year of your master, the course will help you sharpen your thoughts regarding the projects and elective courses you intend to follow. You are invited to motivate your preferred projects for the second semester and your selection of elective courses for the first quarter of the second year.	
Study Goals	Your reflection on your development as offshore & dredging engineer contributes to your proof that you meet the following exit qualifications of the programme: - Self-development: Develop competences and knowledge to function in varying professional environments. - Academic communication: Communicate professional and scientific judgments effectively. - Collaborative problem-solving: Collaborate in international, multi-cultural and multi-disciplinary teams. - Ethical Intelligence: Demonstrate the ability to critically assess the societal and environmental implications of their engineering solutions.	
Education Method	Throughout the first year of the programme you will work on an individual reflection portfolio. In this portfolio, you answer a prepared set of reflective questions for each subsequent quarter. By working on these questions during each quarter, you will complete the learning lines corresponding to the four learning objective of the course. In order to answer the questions, you reflect on your experiences during the courses of each subsequent quarter. Through BrightSpace, we will also provide inspirational materials, such as TedX presentations, YouTube videos and short texts and suggestions for useful books - which can help you during your reflections. You are of course invited to look for your own materials that help you shape your thoughts. Throughout your studies, you are invited to discuss the different topics with the lecturers of your courses and your fellow students.	
Literature and Study Materials	Your participation in the different courses of the programme should provide you with relevant experiences to reflect upon. Supporting study material and references are provided through BrightSpace. In addition, you are invited to explore further material in literature and on the internet.	
Assessment	While reflection can of course not be graded, proving your development is mandatory within the master Offshore & Dredging Engineering. Therefore, you need to submit an individual portfolio at two instances of your studies, at the end of Q2 and at the end of Q4, with the following corresponding deadlines: - Part 1 should be submitted latest on the Friday of week 2.8 - Part 2 should be submitted latest on the Friday of week 4.8 Both parts of the portfolio contain your answers to the eight reflective questions (four questions per quarter). In addition, there is a section in each part with your motivated choices for the following semester. In Part 1, you will motivate your choices regarding project work in the second semester and in Part 2 you will motivate your choices regarding elective courses in the first quarter of the second year. Based on your motivations, we can decide to honour your preferences. The default format of your portfolio is one written page per quarter for answering your questions (A4, font Arial, size 11) and maximum one page with your motivated choices per semester. The size of each part of your portfolio is therefore three pages at max. Nevertheless, you have the freedom to choose an alternative digital format of your portfolio. You could instead decide to submit your reflection as a voice recording, video or by means of a slide show, or any other format that fits your personal style and preference. The duration of any recording should be between 7-10 minutes per quarter. You can submit your portfolio in your personal folder on BrightSpace. Both parts of your reflection portfolio will be assessed as pass/not yet pass. Please know that your reflections as such can never be as 'not yet pass'. A 'not yet pass' would indicate that completeness and ownership of your reflections could be improved. In this case, you will have the opportunity to submit an updated version of your portfolio for second assessment. Based on your motivations, you will be assigned to projects in the second semester and approval of your selected will be decided upon. In addition, you will be invited for a mandatory interview at two instances throughout the year (end of Semesters 1 and 2). Latest one day before the interview, you should submit a draft version of your reflection portfolio. Further information regarding these interviews will be provided through BrightSpace. You can submit your draft portfolio in your personal folder on BrightSpace. Please note that failure to submit shifts the deadline to the next semester (e.g., from week 2.8 to week 4.8). Part 1 should always be submitted one semester before Part 2. Deviation from these regulations requires approval from the Board of Examiners.	
Enrolment / Application	In order to participate in the course, registration on BrightSpace is mandatory.	
Department	ME Department Maritime & Transport Technology	

OE44200	Offshore Development Project	5
Responsible Instructor	Dr.ir. R.L.J. Helmons	
Responsible Instructor	Dr.ir. A. Jarquin Laguna	
Instructor	Dr.ir. B.C. Ummels	
Instructor	Ir. A.C.M. van der Stap	
Contact Hours / Week x/x/x/x	HC 0.0.4.0 W 0.0.2.0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course will introduce the students to the technical aspects and detailed understanding of offshore developments (e.g. renewable energy farms, deep-sea mining, seabed preparations). The course builds on fundamental knowledge addressed in the first semester with project management techniques, as well as applied design methodologies. The students will participate in a realistic project teamwork, working on actual cases with stakeholder analysis, risk management assessment and modelling, including an overview of the project development of safe and reliable offshore technologies. In this course, students can choose from different development projects to participate in (currently two projects available).	
Study Goals	After this course, students should be able to: - Define the design space of a new or modified offshore & dredging engineering development, based on a functional and requirements analysis, and accounting for the full life-cycle - Select design solutions in a methodological and transparent manner, based on an analysis of the life-cycle performance in terms of functional, societal and economical value, including sustainability and circularity - Perform risk analyses, accounting for health, safety and environmental risks, and including possible mitigation measures - Create a plan for professional development, based on individual goals obtained from self-assessment and feedback from peers and supervisors - Articulate assumptions, boundary conditions and limitations of (design) methods, distinguishing concept-based and application-based approaches, to put results, conclusions and recommendations into perspective - Execute well-planned projects, based on principles of project management and time management - Create morally adequate measures for reduction of impact or mitigation of adverse effects	
Education Method	Lectures and project teamwork	
Assessment	Grading based on a group assignment. Portfolio (to be evaluated as a part of the course Reflection for Offshore & Dredging Engineers)	
Department	ME Department Maritime & Transport Technology	

OE44210	Offshore Design Project	5
Responsible Instructor	Dr.ir. P. van der Male	
Responsible Instructor	Ir. J.S. Hoving	
Contact Hours / Week x/x/x/x	HC 0.0.4.0 W 0.0.2.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Successful participation in the following courses (or equivalent) is advised: - OE44150 Marine Hydromechanics - OE44170 Offshore Renewables Technologies - OE44175 Structural Mechanics of Offshore Structures - OE44180 Seabed Installation and Installation Process - OE44205 Offshore Geotechnical Engineering - MT44021 Motions and Loading of Structures in Waves	
Course Contents	The primary objective of this course is to enable students to produce conceptual engineering designs for offshore structures accounting for ultimate strength and fatigue criteria. During this course, students gain a thorough theoretical understanding of the design process, its structuring, available design methodologies and the philosophy behind the design codes. Throughout the course students work on a group design project, where each student team designs an offshore support structure from scratch up to a preliminary design by performing a full offshore support structure design-cycle. This design-cycle includes: I) the assembly of the structural configuration, II) the determination and assessment of the permanent, variable and environmental loads on the structure, III) the design and analysis of its foundation, and IV) the static and dynamic structural analysis of the structure based on ultimate strength and fatigue principles. By means of this group design project, students will learn to judge their design against ULS and FLS design criteria and be able to modify and improve their design based on static and dynamic structural analyses. Thereby, students will be able to analyse and evaluate the dimensioning and performance of structural components for strength and stability. Furthermore, students learn to predict how changing (parts of) their design will influence the corresponding loads and based on their insight in these loads, be able to defend their design choices. Moreover, students will learn what the practical implications of their design choices are for operations and other feasibility and life-cycle aspects. To reinforce the understanding of these aspects, the course program also includes a visit to an offshore structures fabrication yard. During this module, students will not only learn how to design an offshore structure, but they also learn how to collaborate effectively in a team and will further develop skills and competences to motivate their assumptions and provide argumentation of their design choices by accounting for relevant moral values, norms and principles.	
Study Goals	After a successful participation in this course, students are able to: 1. Define the design space of new or modified offshore & dredging engineering structures or operations, based on a functional and requirements analysis, and accounting for the full life-cycle. 2. Select design solutions in a methodological and transparent manner, based on an analysis of the life-cycle performance in terms of functional, societal and economical value, including sustainability and circularity. 3. Perform risk analyses, accounting for health, safety and environmental risks, and including possible mitigation measures. 4. Evaluate the limit states of designs of offshore structures, using an applicable safety philosophy (deterministic, probabilistic) based on valid design standards. 5. Create a plan for professional development, based on individual goals obtained from self-assessment and feedback from peers and supervisors. 6. Articulate assumptions, boundary conditions and limitations of (design) methods, distinguishing concept-based and application-based approaches, to put results, conclusions and recommendations into perspective. 7. Execute well-planned projects, based on principles of project management and time management. This course contributes to the final attainments of the Offshore & Dredging Engineering programme on Constructive Design (1-4), Self-Development (5), Academic Communication (6), Collaborative Problem-Solving (7) and Ethical Intelligence (8).	
Education Method	The education in this course includes problem-based lectures, workshops, a group design project and self-study. In total, this course has 8 contact hours per week, consisting of: - two 2-hour lectures (2x45 min) focused on comprehension and application of theory and methodology of offshore support structures design, and - a 4-hour workshop (4x45 mins) focused on problem analysis and evaluation, treating and discussing example problems, as well as to provide feedback and reflection on the group design project as well as allowing for discussion and Q&A regarding the group design project. The group design project is performed in groups and consists of 3 phases. Each phase comes with a formative assessment and feedback session per group in interaction with the lecturers and/or your peers, followed by a final deadline at the end of the teaching period at which the assignment is graded. The groups generally consist of 3 or 4 students, but the group size may be adapted depending on cohort size. Additionally, under reservation of unforeseen and/or external circumstances, the module includes an excursion to a fabrication yard where offshore support structures are constructed. Here note that the number of participants to this excursion is generally limited by the capacity at the yard and the excursion is therefore not a compulsory part of the module.	
Course Relations	This course is part of the MSc Offshore & Dredging Engineering (ODE) and the Offshore Engineering track of the European Wind Energy Master (EWEM-OE). The course runs parallel and in sync with the unit Design of Offshore Structures, embedded in the module CIEM4120 Offshore Structures, which is part of the track Hydraulic and Offshore Structures (HOS) of the MSc Civil Engineering. The course builds on the courses that are offered in the first and second quarters of the first year of the ODE curriculum.	
Literature and Study Materials	The lecture material for this course consists of the lecture slides, lecture notes, papers and chapters from relevant literature, and a basis for design document for the group design project, which will all be available through Brightspace.	
Assessment	Participants of this module are summatively assessed through a Group Design Project, worth 4EC, and an individual written exam, worth 1EC. The final grade for the course is the weighted average of the Group Design Project (80%) and the individual written exam (20%). To pass the course, participants must score a pass grade, i.e. at least a 6.0 (rounded to halves), on both the project and the exam. In addition, participation in the course allows students to work on an individual reflection portfolio, which is assessed within OE44195 Reflection in Offshore & Dredging Engineering (Study Goals 5-8).	

	Group Design Project: The final deadline for the Group Design Project is set at the end of the Q3 education period. If you fail to get a pass grade for the assignment, but the assignment grade is equal to or higher than a 5.0, you are allowed to improve your grade by submitting an addendum to your assignment report in the following education period, i.e. during Q4. Here, please note that the maximum grade that can be obtained after submitting an addendum to improve an insufficient grade is 6.0. Written exam: The regular individual written exam for this course is organised during the exam period of Q3, and a re-exam is organised in the exam period of Q4. The written exam is an open-book exam, meaning that any written or printed material is permitted during the exam sessions. If due to any unforeseen circumstances the written exam for this course may not be given in a plenary on-campus session, alternatively the written exam for this course may be given as an online open-book exam through Brightspace. To participate in the online exam you must be enrolled for this course in Brightspace. For further practicalities regarding any online examinations, you will be informed through Brightspace.
Permitted Materials during Tests	The written exam is an open-book exam, meaning that any books, slide print-outs, personal notes, i.e. any written or printed material is permitted during the exam sessions. However, to limit the amount of paper waste, a maximum of ten A4-pages (written or printed) are allowed during the exam. This rule does not include books. Additionally, it is allowed and advised to bring a scientific calculator to the exam. Any devices that allow for communication are not allowed, i.e. no laptops, pads or phones.
Enrolment / Application	For this course, enrollment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations.
Remarks	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/regulations. Through the past years, this course has seen several updates and improvements, and thereby updated course codes. Previous course codes are: OE4651, OE4651-12, OE4651-15, OE44095, OE44096 and OE44097.

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE Semester 3 NTNU

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSC ODE Semester 4 TU Delft or NTNU

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Offshore and Dredging Engineering

MSC ODE Semester 4 TU Delft or NTNU Graduation Project

OE54015EWEM	EWEM Literature Research	15
Responsible Instructor	Dr.ir. P. van der Male	
Course Contents	This course code serves as a place holder for the specialization project of EWEM students of the Offshore Engineering track (EWEM-OE), which they will execute at NTNU. For more details, please contact the EWEM-OE track coordinator at NTNU.	

OE54030EWEM	EWEM MSc Thesis	30
Responsible Instructor	Dr.ir. P. van der Male	
Course Contents	This course concerns the MSc thesis project for EWEM-students of the Offshore Engineering track (EWEM-OE), and is similar to the course OE54035. For more information, contact the EWEM-OE track coordinators at TU Delft and NTNU.	

Dr.ir. S.M.S. Alhaddad

Department	Offshore and Dredging Engineering
Telephone	+31 15 27 83675
Room	34.B-2-230

Dr. K. Anupam

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 82394
Room	23.HG 6.60

Dr.ir. J.H. den Besten

Unit	Mech, Maritime & Materials Eng
Department	Ship and Offshore Structures
Telephone	+31 15 27 86507
Room	34.B-1-290

Dr.ir. A.J.J. van den Boom

Unit	Mech, Maritime & Materials Eng
Department	Team Ton van den Boom
Telephone	+31 15 27 84052
Room	34.C-3-180

Ir. B.J.E. de Bruin

Department	Cognitive Robotics
Department	Cognitive Robotics

Dr. H. Caesar

Department	Intelligent Vehicles
Telephone	+31 15 27 82419

J.O. Colomes Gene

Department	Offshore Engineering
Telephone	+31 15 27 83461
Room	23.HG 3.65

Prof.dr.ir. B.H.K. De Schutter

Unit	Mech, Maritime & Materials Eng
Department	Delft Center for Systems and Control
Telephone	+31 15 27 85113

D. Fisaletti

Department	Ship Hydromechanics
Telephone	+31 15 27 81697
Department	Wind Energy
Telephone	+31 15 27 81697

Dr. A. Grammatikopoulos

Department	Ship and Offshore Structures
-------------------	------------------------------

Prof.dr.ir. J. Hellendoorn

Department	Cognitive Robotics
Telephone	+31 15 27 89007
Room	34.F-2-440
Unit	Mech, Maritime & Materials Eng
Department	College van Bestuur
Telephone	+31 15 27 89007
Room	34.F-2-440

Dr.ir. R.L.J. Helmons

Unit	Mech, Maritime & Materials Eng
Department	Offshore and Dredging Engineering
Telephone	+31 15 27 84746

Dr.ir. H. Hendrikse

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 88223
Room	23.HG 3.71

Y. de las Heras

Unit	Civiele Techniek & Geowetensch
Department	Education and Student Affairs CEG
Telephone	+31 15 27 84800
Room	23.HG 2.73

Ir. J.S. Hoving

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 85723
Room	23.HG 3.61.1

Dr.ir. A. Jarquin Laguna

Unit	Civiele Techniek & Geowetensch
Department	Offshore and Dredging Engineering
Telephone	+31 15 27 83297

Dr.ir. G.H. Keetels

Unit	Mech, Maritime & Materials Eng
Department	Offshore and Dredging Engineering
Telephone	+31 15 27 84057

Dr. E. Kementzetidis

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Room	23.HG 3.45

Prof.dr.ir. T. Keviczky

Unit	Mech, Maritime & Materials Eng
Department	Team Tamas Keviczky
Telephone	+31 15 27 82928
Room	34.C-3-310

H. Khodaei

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 82759
Room	31.C2.130

Dr.ir. J. Kober

Unit	Mech, Maritime & Materials Eng
Department	Learning & Autonomous Control
Telephone	+31 15 27 85150
Room	34.F-2-380

Dr. G. Lavidas

Department	Offshore Engineering
Telephone	+31 15 27 83864

Dr.ir. P. van der Male

Unit	Civiele Techniek & Geowetensch
Department	Ship Hydromechanics and Structures

Telephone Room	+31 15 27 83473 23.HG 3.71
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Prof.dr. A.V. Metrikine

Unit Department	Civiele Techniek & Geowetensch Offshore Engineering
Telephone Room	+31 15 27 84749 23.HG 3.59.1

Unit Department	Civiele Techniek & Geowetensch Dynamics of Structures
Telephone Room	+31 15 27 84749 23.HG 3.59.1

Dr.ir. P. Naaijen

Unit Department	Mech, Maritime & Materials Eng Ship Hydromechanics and Structures
Telephone	+31 15 27 81570

G.O. Ndubuisi

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 88895

L. Pahlavan

Unit Department	Mech, Maritime & Materials Eng Ship and Offshore Structures
Telephone	+31 15 27 87682

L. Peternel

Unit Department	Mech, Maritime & Materials Eng Human-Robot Interaction
Telephone Room	+31 15 27 85695 34.F-1-420

J.M. Prendergast

Department	Human-Robot Interaction
Telephone	+31 15 27 87944

J. Schmutzler

Department	Delft Centre for Entrepreneurship
Department	Delft Centre for Entrepreneurship

Dr.ing. V.E. Scholten

Unit Department	Techniek, Bestuur & Management Delft Centre for Entrepreneurship
Telephone Room	+31 15 27 89596 31.C2.070

B. Shyrokau

Unit Department	Mech, Maritime & Materials Eng Intelligent Vehicles
Telephone Room	+31 15 27 84536 34.E-0-220

Ir. A.C.M. van der Stap

Unit Department	Civiele Techniek & Geowetensch Offshore Engineering
Room	23.HG 3.48

Dr. S.T.H. Storm

Unit	Techniek, Bestuur & Management
-------------	--------------------------------

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 83548
Room	31.c2.160

P. Taneja

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Telephone	+31 15 27 85438
Room	23.HG 3.86

Dr.ir. B.C. Ummels

Department	Offshore Engineering
Room	23.HG 3.36

Prof.dr.ir. L.J. de Vries

Unit	Techniek, Bestuur & Management
Department	Energie en Industrie
Telephone	+31 15 27 81137
Room	31.B3.030

Dr.ir. S. de Vries

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 89220
Room	23.HG 3.70

Dr. C.L. Walters

Unit	Mech, Maritime & Materials Eng
Department	Ship and Offshore Structures
Telephone	+31 15 27 83705
Room	34.B-1-310

Ir. J.W. Welleman

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 84856
Room	23.HG 6.55

Dr.ir. P.R. Wellens

Unit	Mech, Maritime & Materials Eng
Department	Ship Hydromechanics and Structures
Telephone	+31 15 27 88251

Dr. D. Wüthrich

Department	Hydraulic Structures and Flood Risk
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